ISMIR 2026 Tutorial

Music for Learning:

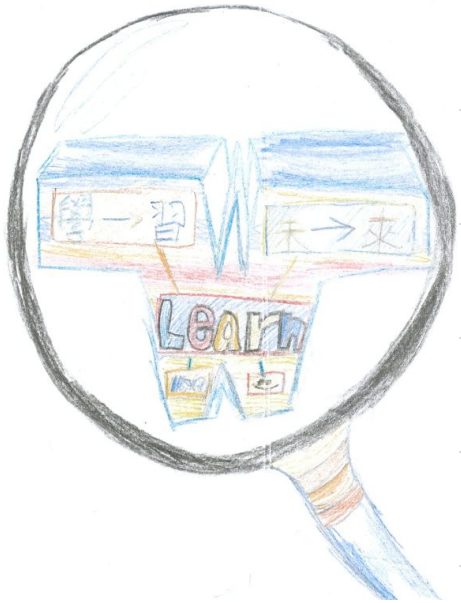
From Experimental Design to Multimodal Measurement

Ying Que, Assistant Professor, School of Psychology, South China Normal University
Xiao Hu, Associate Professor, College of Information Science, The University of Arizona
Fanjie Li, PhD candidate, Department of Teaching and Learning, Vanderbilt University
Ruilun Liu, PhD, Faculty of Education, The University of Hong Kong

Abu Dhabi · November 8, 2026 · 14:00–17:30

Slides (PDF): bit.ly/ISMIR26_M4L Padlet for interaction: bit.ly/M4L_Padlet

Agenda



Time	Session	Presenter
14:00–14:10	1 Welcome & Self-introduction	Xiao Hu
14:10–14:40	2 Foundations	Xiao Hu
14:40–15:20	3 Laboratory experiments	Ying Que
15:20–16:00	4 Naturalistic studies	Fanjie Li & Xiao Hu
16:00–16:15	Coffee break	—
16:15–16:45	5 User-centered measurement	Ying Que
16:45–17:15	6 Hands-on demo	Ying Que & Fanjie Li
17:15–17:30	7 Closing discussion	Xiao Hu, Ying Que, Fanjie Li

Objectives

- To introduce a **methodological framework** for human-centered research on music for learning
- To demonstrate how **music representations** produced by MIR techniques can be **integrated with behavioral, subjective, and physiological measurements**
- To investigate **learners' cognitive processes and affective responses** to music
- To derive **practical considerations** for experimental design, multimodal data collection, measurement, and modelling

Cultural Computing and Multimodal Information Research (CCMIR)


Xiao Hu


Ying Que


Fanjie Li


Ruilun Liu


Zuo Wang


Shen Ba


Jeremy Ng


Nora Hernandez

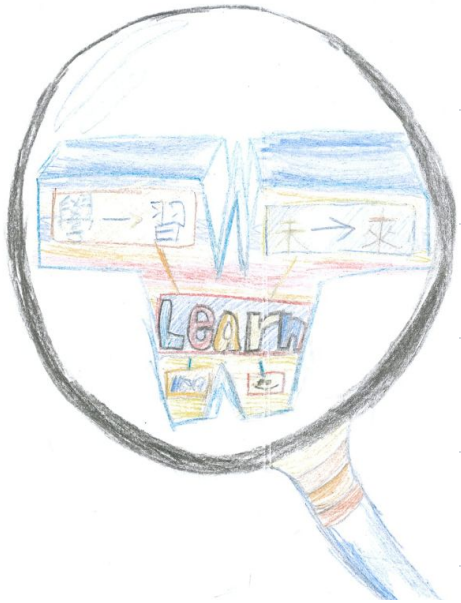
https://ccmirlab.com/

https://www.researchgate.net/lab/CCMIR-Xiao-Hu

国家自然科学基金委员会
National Natural Science Foundation of China

大學教育資助委員會
University Grants Committee

Agenda



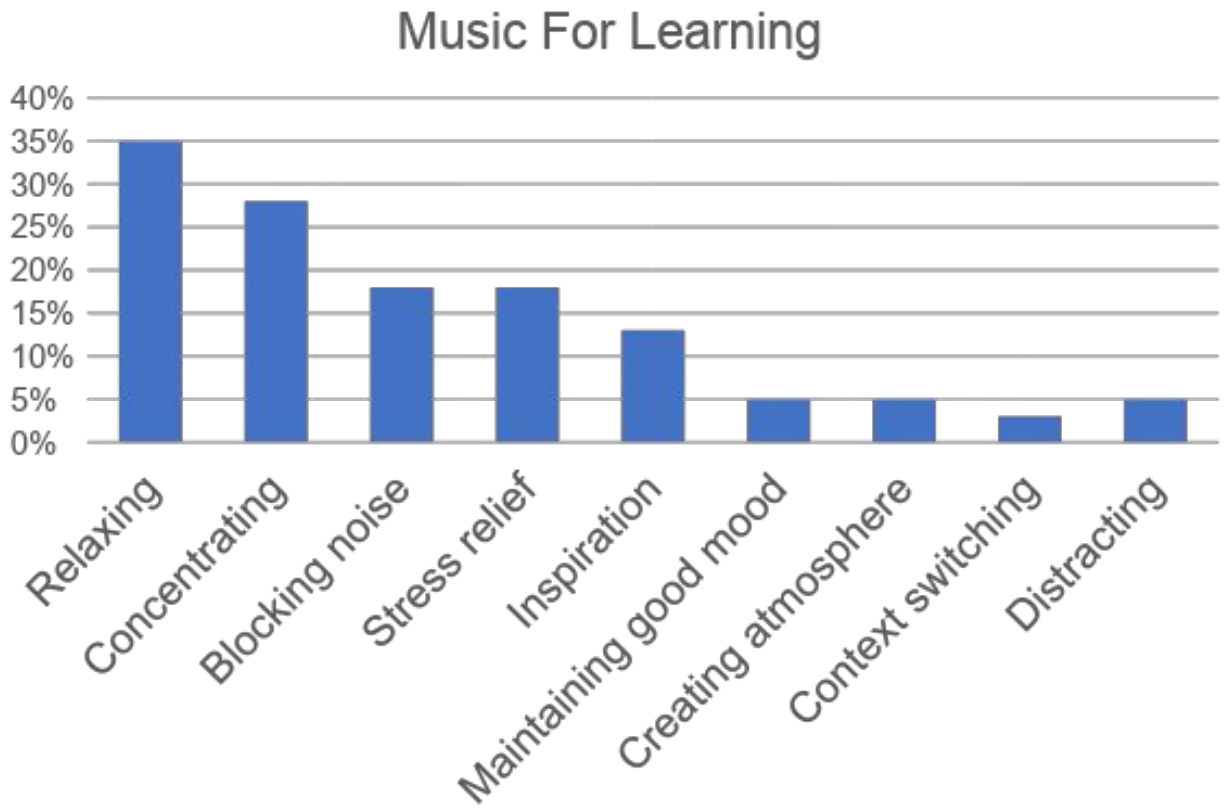
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Music for Learning
&
Connections with MIR

Music Usage in Real Life

- Background Music (BGM) is a common practice (Hu, Chen & Wang, 2021)
  - To adjust mood, behaviour and engagement during learning
  - 2/3 of students played BGM while study, 1/5 of them played during 90% of the learning sessions (survey: Calderwood et al., 2014)



Hu, X., Chen, J., & Wang, Y. (2021). University students' use of music for learning and well-being: A qualitative study and design implications. *Information Processing and Management*. 58(1), 102409.

Music Emotion in MIR

- Perceived Emotion vs. Induced Emotion (Hu, 2010)
- Music Emotion Recognition (MER, Tutorial ISMIR'12)
- Audio Music Mood Classification in MIREX (2007 - 2020)
- Audio Tag Classification task (Mood tag dataset) in MIREX (2009 - 2021)

- Cross-cultural Music Mood Classification (Hu & Lee, 2016; Hu & Yang, 2017)
- Cross-cultural studies in MIR (WiMIR special session, ISMIR'21)

- Physiological signal based MER (Hu, Li & Ng, ISMIR'18; Liu & Hu, 2020; Que & Hu, 2025)
- Personalized and context-sensitive MER (Gómez-Cañón et al., 2021)

Multimodality in MIR

- Physiological signal based MER (Hu, Li & Ng, ISMIR'18; Liu & Hu, 2020; Que & Hu, 2025)
- Personalized and context-sensitive MER (Gómez-Cañón et al., 2021)

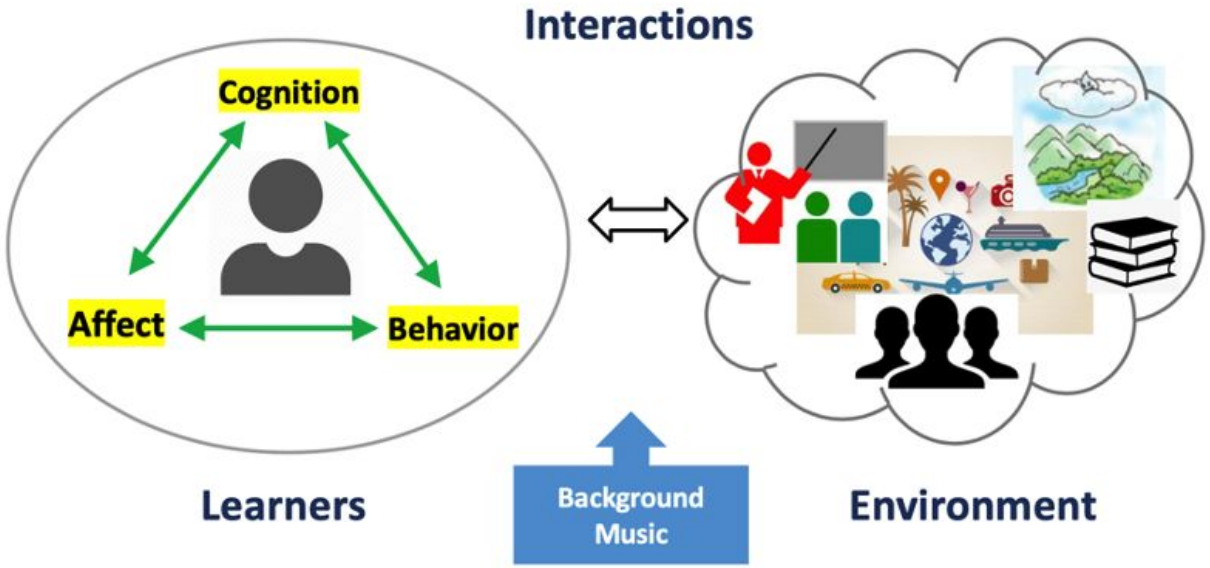
- EEG decoding for MIR research (Tutorial in ISMIR'16, by Stober et al.)
- Neural Annotation Dataset Study (ISMIR'17 by Stober et al.)
- Analysing physiological data collected during music listening (Tutorial in ISMIR'23 by Bishop et al.)
- ISMIR late-breaking demos and sessions integrating brain signals with generative audio models, such as work on EEG-token-based music generation (Cho, 2025)

Theoretical Background

- Music as an accompaniment to learning activities & a tool for emotion modulation (arousal-mood hypothesis)
- Emotion affects learning (control-value theory)
- Music causes cognitive load during learning (cognitive load theory)
- Inconsistent findings on the effects of music on learning call for new evidence
  - Most studies were based on behavioral measures, could multimodal learning analytics help?
  - Potential influential factors: Inadequate considerations of music, learner, task characteristics

Theoretical Background (Con't)

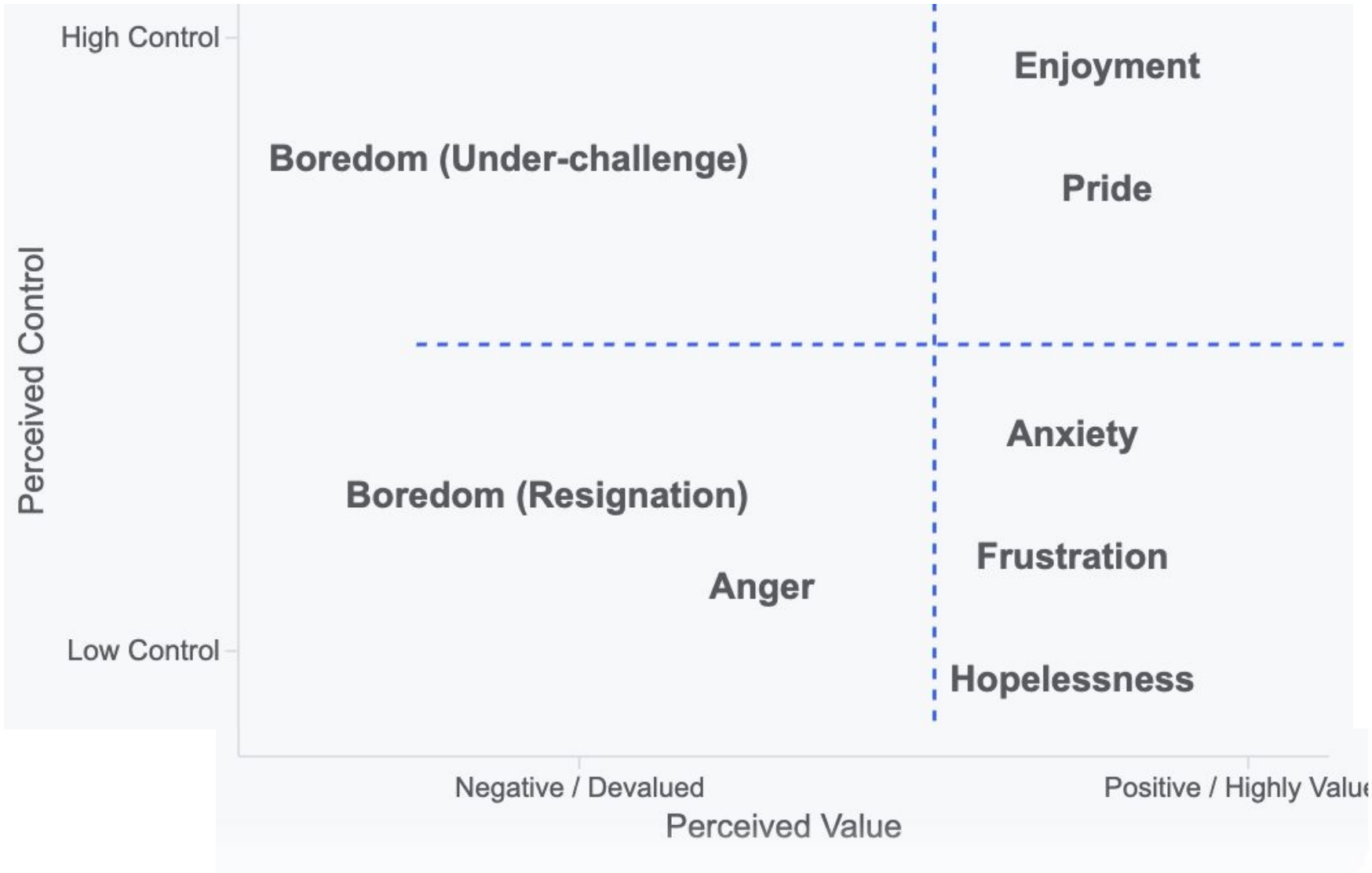
- ABC model of social psychology:
  - Human beings rely on basic capacities of Affect, Behaviour and Cognition to successfully interact with the environment (Allport, 1985; Jhangiani et al., 2014).
- Our research:
  - Propose to probe the effect of BGM on learning from perspectives of affect, behaviour and cognition.



Theoretical Background: Affect

- Arousal and Mood Hypothesis (Husain et al., 2002)
  - It indicates that music influences learning by affecting the arousal and mood.
  - Arousal represents degree of energy, mood represents degree of pleasantness.
- Control-value theory of achievement emotions (Pekrun & Raimond, 2014)
  - Affective qualities are directly associated with the accomplishment of learning activities.
  - Achievement emotions (that arise during a task) are shaped by appraisals on control and value
  - It is generally believed that positive emotions are beneficial to the learning process whereas negative emotions are detrimental.

Control-Value Achievement Emotion



Theoretical Background: Behavior

- Alter Task Engagement and Performance
  - BGM could be a non-intrusive & effective strategy to facilitate **task engagement** (Schwartz et al., 2017) .
  - Active engagement in academic tasks could lead to better **learning outcome** (Hallam & MacDonald, 2009).
  - BGM is helpful to **decrease stimulatory and aberrant behavior** (Ahearn et al., 2007).
  - BGM could **distract students** (Thompson et al., 2012), which in turn, impair learning outcome.

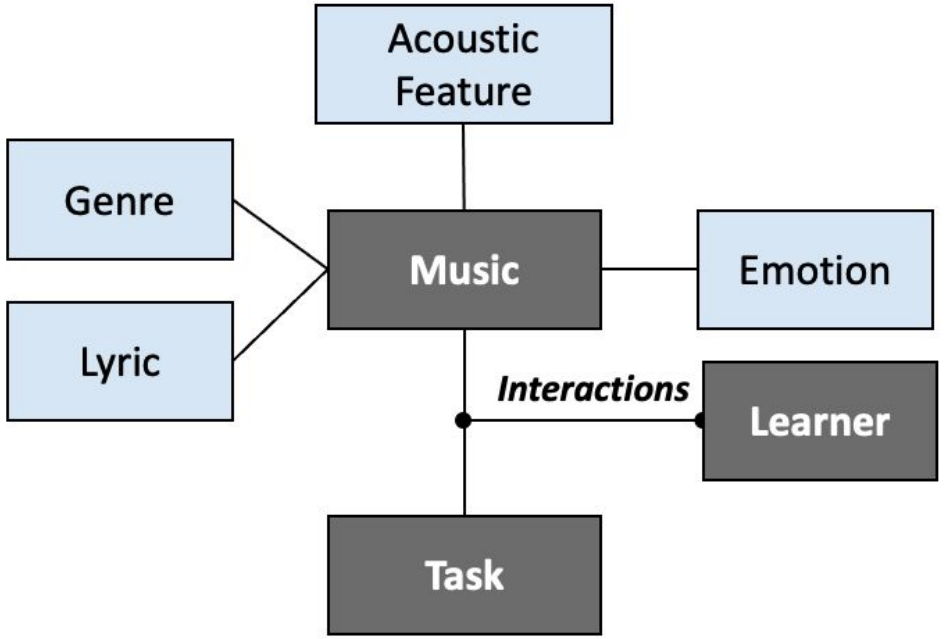
Theoretical Background: Cognition

- Seductive Detail Effect
  - Auditory reception is an inherent function in human's brain
  - BGM consumes brain's cognitive capacity, which may draw learners' attention from the mainly executing task (Rey, 2012).
  - BGM would increase cognitive load, and thereof, impede learning outcome (Perham & Currie, 2014).
- Combination
  - Music affect learning outcomes through changing learners' emotion

Ba, S. & Hu, X. (2025). Effects of background music tempo and mode on reading comprehension: The mediating role of emotions. *Instructional Science*. 1-23. https://doi.org/10.1007/s11251-025-09728-5

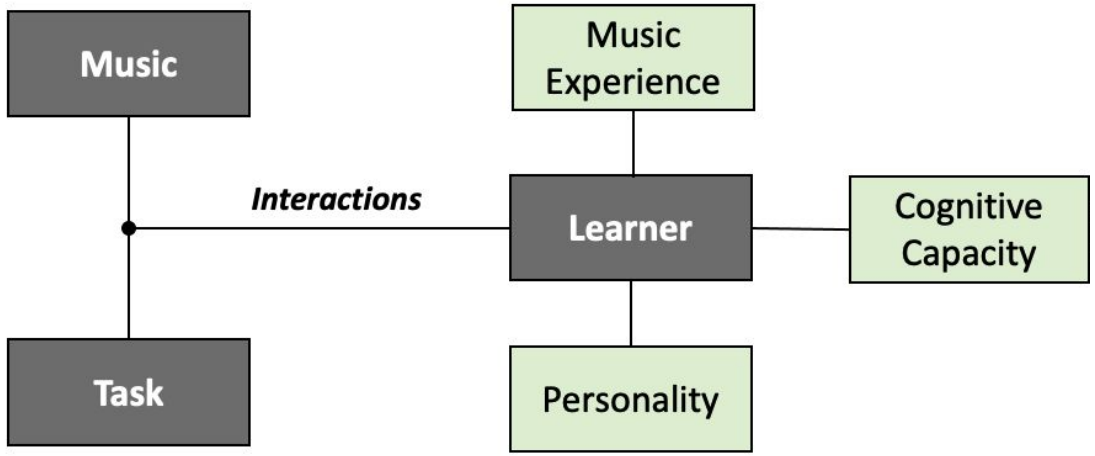
Interplay among Music, Learner, Task

- Music Characteristics
  - **Genre:** "hip-hop" music more distracting than "light classical" music which was more intrusive than "silence" (Chou, 2010).
  - **Acoustic Features**: fast-paced & high-intensity BGM interfered reading (Thompson, 2012).
  - **Lyric**: BGM without lyrics linked to higher reading scores despite at insignificant level (Liapis et al., 2008).
  - **Emotion**: Musical emotions (i.e., happiness & energy) positively correlated to task engagement (Li et al., 2020).



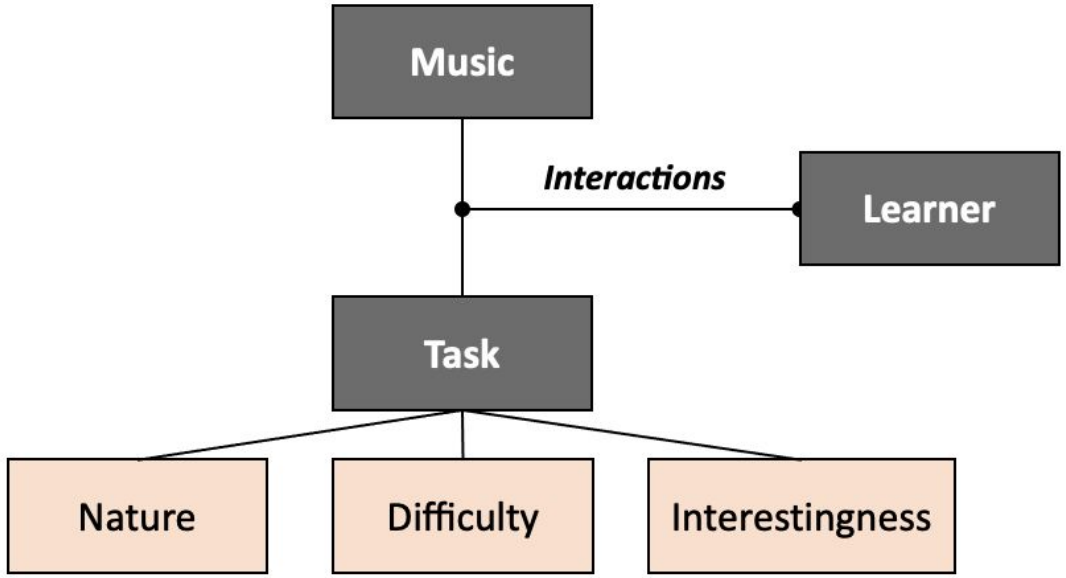
Interplay among Music, Learner, Task

- Learner Characteristics
  - **Personality Traits:** Introverts more likely to suffer from BGM due to their relatively higher "cortical arousal" than extraverts (Küsner, 2017).
  - **Music Experience:** Learners used to listening to BM during study fulfilled the verbal learning tasks better than those not (Etaugh & Ptasnik, 1982).
  - **Cognitive Capacity:** BGM occupied limited working memory resources preventing learners' fully participating in task (Lehmann & Seufert, 2017).



Interplay among Music, Learner, Task

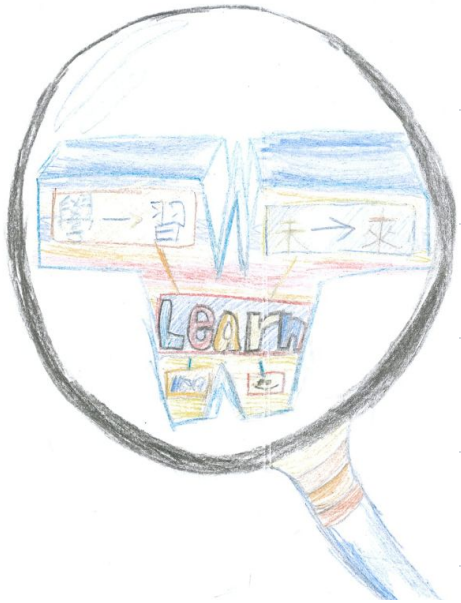
- Task Characteristics
  - Task **nature, difficulty level, interestingness** might moderate the effect of BGM on learning (Kämpfe et al., 2011; Hallam, 2010; Lehmann & Seufert, 2017).



General Approaches

- Experiment: Informed by neuroscience and psychology
  - Lab vs. field
  - Between subjects vs. Within subjects
  - Tasks: Reading comprehension vs. Virtual reality appreciation vs. Self-chosen tasks
  - Music: researcher selected vs. self-selected vs. AI recommended
- Multimodal and user-centered measures
  - Personal traits
  - Behavioral
  - Perceptual
  - Physiological, facial expression, eye movement, brain signals

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Pros and Cons of Lab Experiment

- Better control of experiment environments
- Better control of confounding variables
- Use of expensive and bulky devices
- Higher internal validity (better for causal inference)

Therefore, lab experiments are often more complex, and adopted before naturalistic experiments

- Lower external validity (real-world generalizability)
- Higher participation bias

Music, Users, and Tasks

The following studies used various Background Music

- Tempo: Fast vs. Slow; Mode: Major vs. Minor
- Self-chosen by users
- Selected to match visual stimuli

Users: University students w/ varied

- Music preference
- Personality
- Cognitive capacity

Tasks: Visual learning tasks (to complement with audio channel)

- Reading comprehension (text passages)
- Immersive scene appreciation (virtual reality)

Study 1: Behavioral + Physiological Data + Facial Expression (N = 33)



	4 Music Pieces		1 Environmental Sound
TEMPO	fast	slow	
MODE	major	minor	

Music/Audio played continuously

Question set 1		Question set 2	Reading	Question set 3	Q/A		
emotional status	Music / Audio (1 min)	emotional status and music/audio	Reading passage	emotional status and metacognition	T/F questions	Break (20s)	Exit Interview

- Valence
- Arousal

- Valence, Arousal
- Familiarity, Enjoyment (with music/audio)

- Valence, Arousal
- Familiarity, Understanding, Perceived difficulty, Confidence of answering questions (about passage)
- Engagement (with reading)

Hu, X., Li, F. & Kong, R. (2019). Can background music facilitate learning? Preliminary results on reading comprehension. In *Proceedings of* the *9th International Conference on Learning Analytics and Knowledge (LAK)*

Physiological Data

Empatica E4



- Electrodermal Activity (EDA)
- Blood Volume Pulse (BVP)
- Inter Beat Interval (IBI)
- Heart Rate (HR)
- Skin Temperature (TEMP)

Indicators Extracted from Physiological Data

CATEGORIES	INDICATORS
Descriptive statistics	Mean, standard deviation, median, range
Time series	Means of the values of the 1st (MFD)/ 2nd (MSD) differences of the signals
Spectrum analysis	High frequency (HF), low frequency (LF), LF / HF

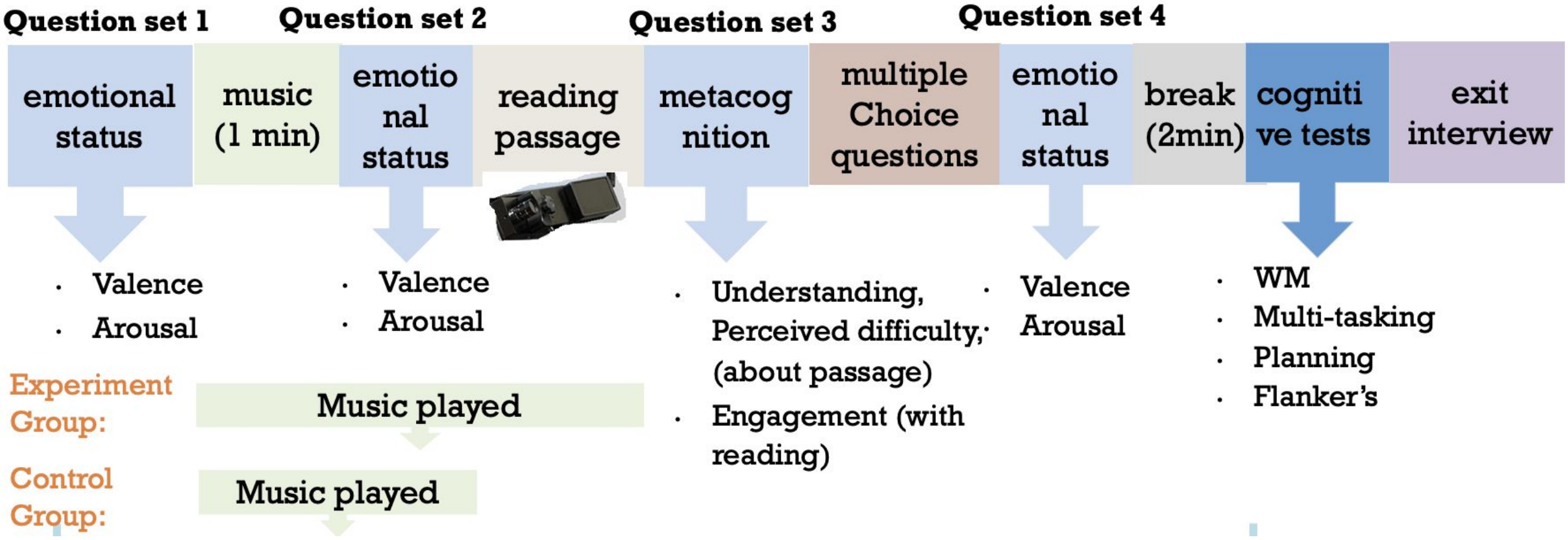
Hu, X., Li, F. & Kong, *R.* (2019). Can background music facilitate learning? Preliminary results on reading comprehension. In *Proceedings of the 9th International Conference on Learning Analytics and Knowledge (LAK)*

Data Analysis & Results

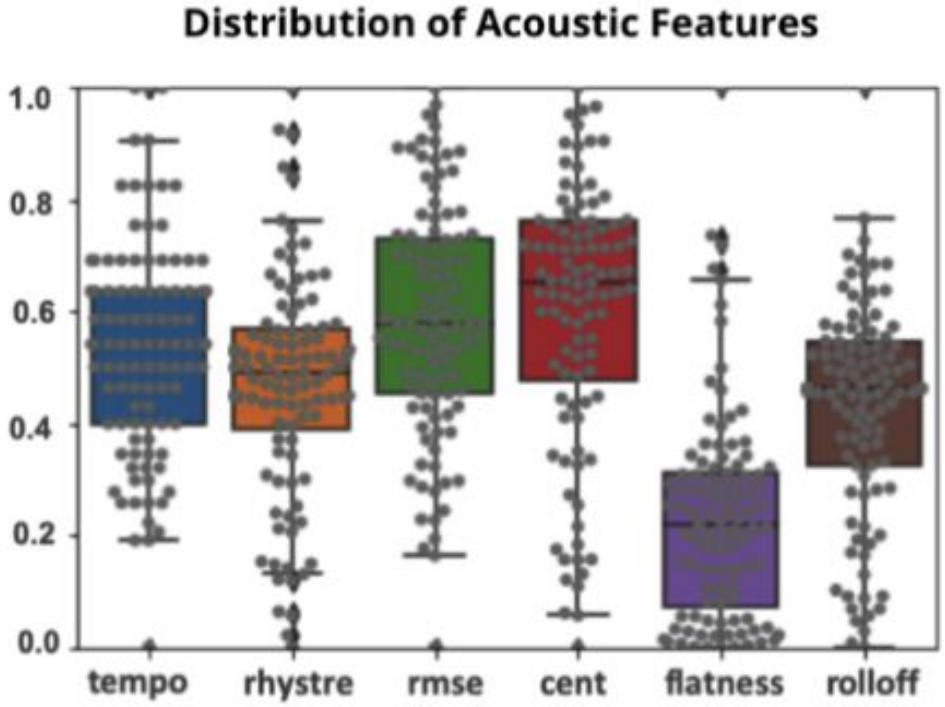
- Comparison across conditions; Linear mixed models
  - Reading performance
  - Emotion (self-reported, facial expression)
  - Physiological measures
- Results
  - Different music had no direct effect on reading performance
  - Music-induced emotions measured by self-reported valence, body temperature, and heart rate were significantly associated with reading comprehension performance

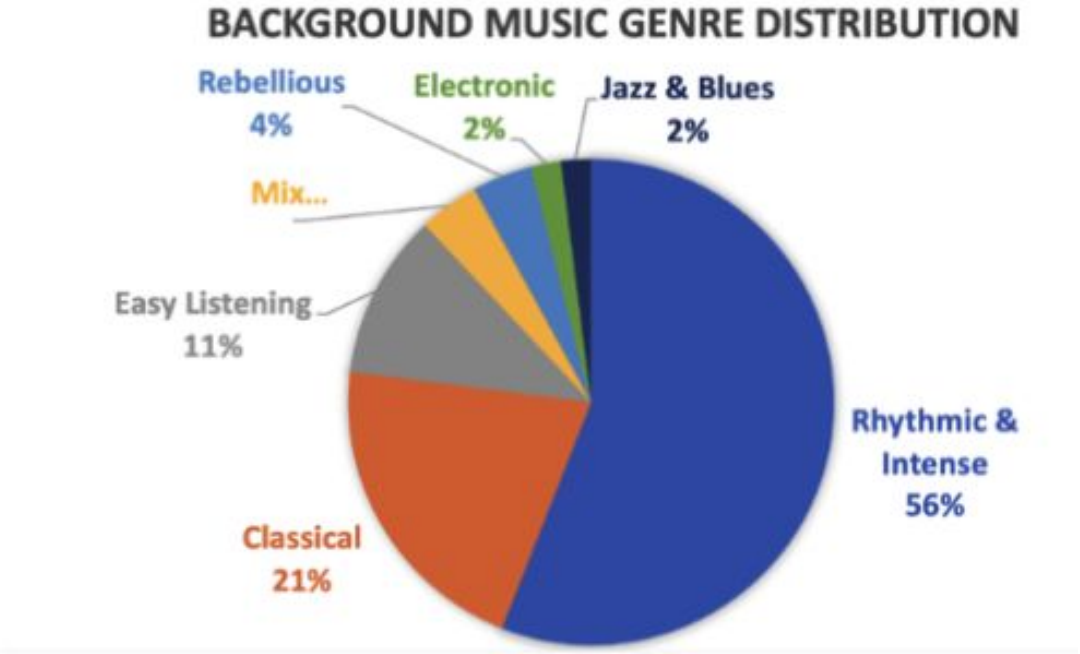
Ba, S., & Hu, X. (2025). Effects of background music tempo and mode on reading comprehension: the mediating role of emotions. *Instructional Science*, 53(5), 1071-*1093*.

Study 2: Behavioral + Physiological + Eye Movement (N = 102; between-subject)



Which Music?
(Methods from MIR)



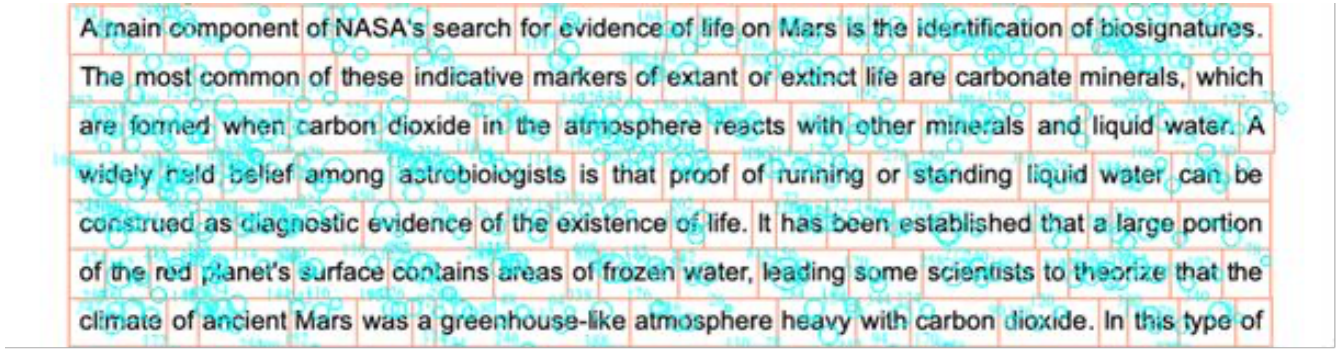


2/3 playlists with lyrics

Que, Y., Zheng, Y., Hsiao, J. H., & Hu, X. (2020, August). Exploring the effect of personalized background music on reading comprehension. In *Proceedings of the ACM/IEEE Joint Conference on Digital Libraries* (JCDL) (pp. 57-66).

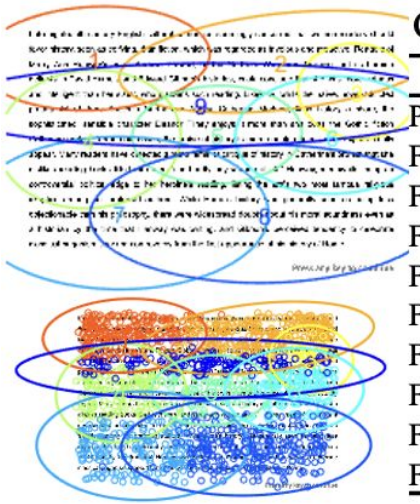
Eye Movement Measures

- Lexical processing stage --- No effect of BGM
  - First Fixation Duration
  - First Pass Dwell Time
  - First Pass Saccade Amplitude
  - First Pass Dwell Time before Skipped Words
- Post-lexical processing stage --- BGM group had higher values
  - Total Fixation Count
  - Total Fixation Duration
  - Total Regression Count
  - Total Word Skipping Rate



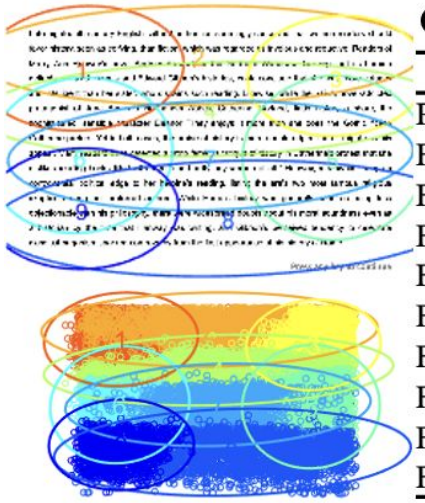
Que, Y., Zheng, Y., Hsiao, J. & Hu, X. (2023). Studying the Effect of Self-selected Background Music on Reading Comprehension with Eye Movements. *Scientific Report, 13(1), 1704.*

Eye Movement Analysis with Hidden Markov Models (EMHMM)



Group A Pattern (5 participants)

	To 1	To 2	To 3	To 4	To 5	To 6	To 7	To 8	To 9
Priors	.52	.11	.02	.06	.04	.05	.06	.10	.04
From 1	.83	.16	.00	.01	.00	.00	.00	.00	.00
From 2	.08	.88	.03	.00	.00	.00	.00	.00	.01
From 3	.16	.01	.72	.11	.00	.00	.00	.00	.00
From 4	.00	.00	.02	.77	.12	.02	.04	.01	.01
From 5	.01	.01	.00	.00	.79	.18	.00	.00	.01
From 6	.00	.00	.00	.16	.00	.80	.02	.00	.00
From 7	.01	.00	.00	.01	.00	.00	.81	.16	.00
From 8	.01	.00	.00	.00	.00	.01	.09	.88	.00
From 9	.02	.00	.00	.00	.00	.01	.00	.02	.94



Group B Pattern (44 participants)

	To 1	To 2	To 3	To 4	To 5	To 6	To 7	To 8	To 9
Priors	.37	.25	.05	.06	.05	.03	.08	.08	.03
From 1	.79	.15	.01	.04	.00	.00	.01	.00	.00
From 2	.01	.92	.06	.00	.00	.00	.00	.00	.00
From 3	.15	.01	.80	.02	.01	.01	.00	.00	.00
From 4	.00	.00	.03	.93	.01	.00	.01	.00	.00
From 5	.00	.00	.00	.00	.78	.13	.01	.01	.06
From 6	.00	.00	.00	.03	.01	.74	.19	.02	.00
From 7	.00	.00	.00	.00	.07	.00	.90	.01	.00
From 8	.00	.00	.00	.00	.02	.00	.01	.94	.02
From 9	.00	.00	.00	.00	.00	.00	.02	.15	.81

Zheng, Y., Que, Y., Hu, X. & Hsiao, J. (2022). Predicting reading performance based on eye movement analysis with hidden Markov models. In *Proceedings of the 22nd IEEE International Conference on Advanced Learning Technologies (ICALT)* (Best Paper Award nomination)

Predict Different Types of Cognitive Load

- logistic regression
- More fixations → higher odds of the BGM condition; additional processing demands from BGM.
- Lower mean HR → higher odds of the BGM condition; relaxation associated with preferred music.
- Processing demands and relaxation coexist.

Targets	Predictors	B (S.E.)	OR [95% CI]	Sig	R^2
Extraneous: Silence (0) vs. BGM (1)	Fixation count	0.002 (0.000)***	1.002 [1.001, 1.003]	0.000	0.082***
	Mean HR	− 0.034 (0.007)***	0.966 [0.954, 0.979]	0.000	
	Constant	1.925 (0.510)***	6.858 [NA, NA]	0.000	
Intrinsic: Easy (0) vs. Hard (1)	Mean saccade amplitude	0.454 (0.099)***	1.575 [1.296, 1.913]	0.000	0.072***
	Fixation Count	0.002 (0.001)***	1.002 [1.001, 1.003]	0.000	
	Constant	− 3.070 (0.608)***	0.046 [NA, NA]	0.000	
Germane: Low (0) vs. High (1)	Fixation Count	0.001 (0.000)**	1.001 [1.000, 1.002]	0.002	0.021**
	HRV RMSSD	0.008 (0.004)*	1.008 [1.001, 1.015]	0.033	
	Constant	− 0.761 (0.264)**	0.467 [NA, NA]	0.004	

Que, Y., Zheng, Y., Hsiao, J. H., & Hu, X. (2025). Using eye movements, electrodermal activities, and heart rates to predict different types *of* cognitive load during reading with background music. *Scientific Reports*, 15(1), 32635.

To what extent can eye movements predict learners' metacognition, in addition to their personal traits, the **characteristics of music**, and text complexity during reading with BGM?

Data analysis:

- Students who read with BGM (N=51); linear regression models

Results:

- Learners' self-reported understanding was positively predicted by working memory capacity and mean saccade amplitude, whereas **negatively predicted** by average fixation duration, **the presence of lyrics** and text complexity.
- Learners' self-reported engagement was positively predicted by regression count, while **negatively predicted by music loudness**, and text complexity.

Que, Y., Zheng, Y., Hsiao, J. H., & Hu, X. (2024, July). Predicting Learners' Meta-cognition Using Eye Movements during Reading with Background Music. In *2024 IEEE International Conference on Advanced Learning Technologies*.

How do learners' self-reported emotions, peripheral physiological signals, and pupillary responses change when they listen to different genres of BGM?

Constructs	Measures	F value	Post hoc Tests between music conditions
Self-report	valence chg	11.27***	silence (-1.68) vs. classical (-0.72) ***
			silence (-1.68) vs. easylisten (-0.66) ***
	arousal chg	5.81***	silence (-1.05) vs. rhythmic (-0.36) ***
EDA	SCR freq chg	29.52***	silence (0.70) vs. classical (-2.26) ***
			silence (0.70) vs. easylisten (-1.04) ***
			silence (0.70) vs. rebellious (-1.28) ***
			silence (0.70) vs. rhythmic (-1.18) ***
	SCR ampltd chg	8.99***	silence (-0.10) vs. rhythmic (-2.66) ***
BVP	Mean BVP chg	7.13***	silence (-0.01) vs. rhythmic (-0.14) ***

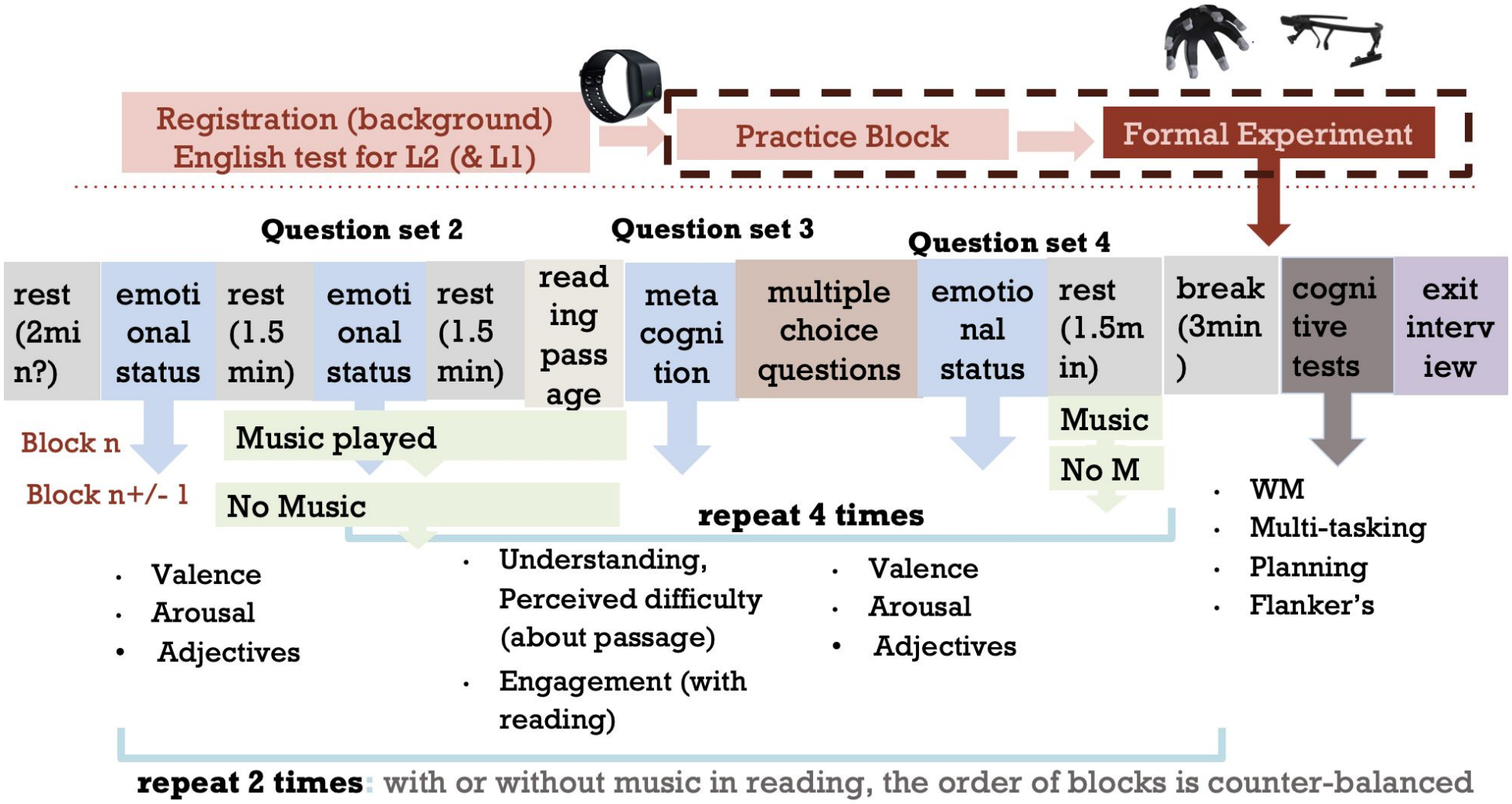
Que, Y., Ng, J. T. D., Hu, X., Mak, M. K. F., & Yip, P. T. Y. (2024, March). Using multimodal learning analytics to examine learners' responses to different types of background music during reading comprehension. In *Proceedings of the 14th Learning Analytics and Knowledge Conference*.

Which BGM characteristics can predict changes in learners' self-reported emotions, peripheral physiological signals, and pupillary responses?

Constructs	Measures	Predictor(s)	Unstd. Coeff. B	Std. Error	Coeff. Beta	t	Sig.	R^2
Self-report	valence chg	rmse	-20.84	5.15	-0.21	-4.04	0.000	.096
		instrumentation	-0.52	0.18	-0.15	-2.94	0.003	
		tempo	-0.01	0.00	-0.14	-2.94	0.003	
	arousal chg	instrumentation	-0.34	0.16	-0.10	-2.22	0.027	.011
Eye-track	pupil size chg	rhythm strength	0.19	0.03	0.26	5.59	0.000	.065
HR	HR mean chg	instrumentation	2.00	0.59	0.17	3.39	0.001	.027
EDA	SCR freq chg	flatness	48.76	10.90	0.22	4.48	0.000	.046

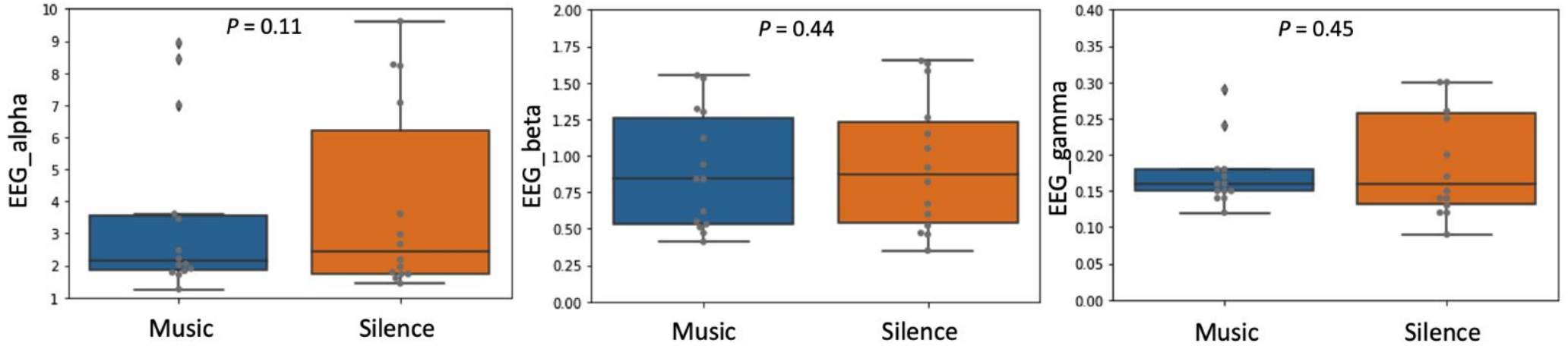
Que, Y., Ng, J. T. D., Hu, X., Mak, M. K. F., & Yip, P. T. Y. (2024, March). Using multimodal learning analytics to examine learners' responses to different types of background music during reading comprehension. In *Proceedings of the 14th Learning Analytics and Knowledge Conference*.

Study 3-1: Behavioral + Physiological + Eye movement + Brain signals (N = 14)



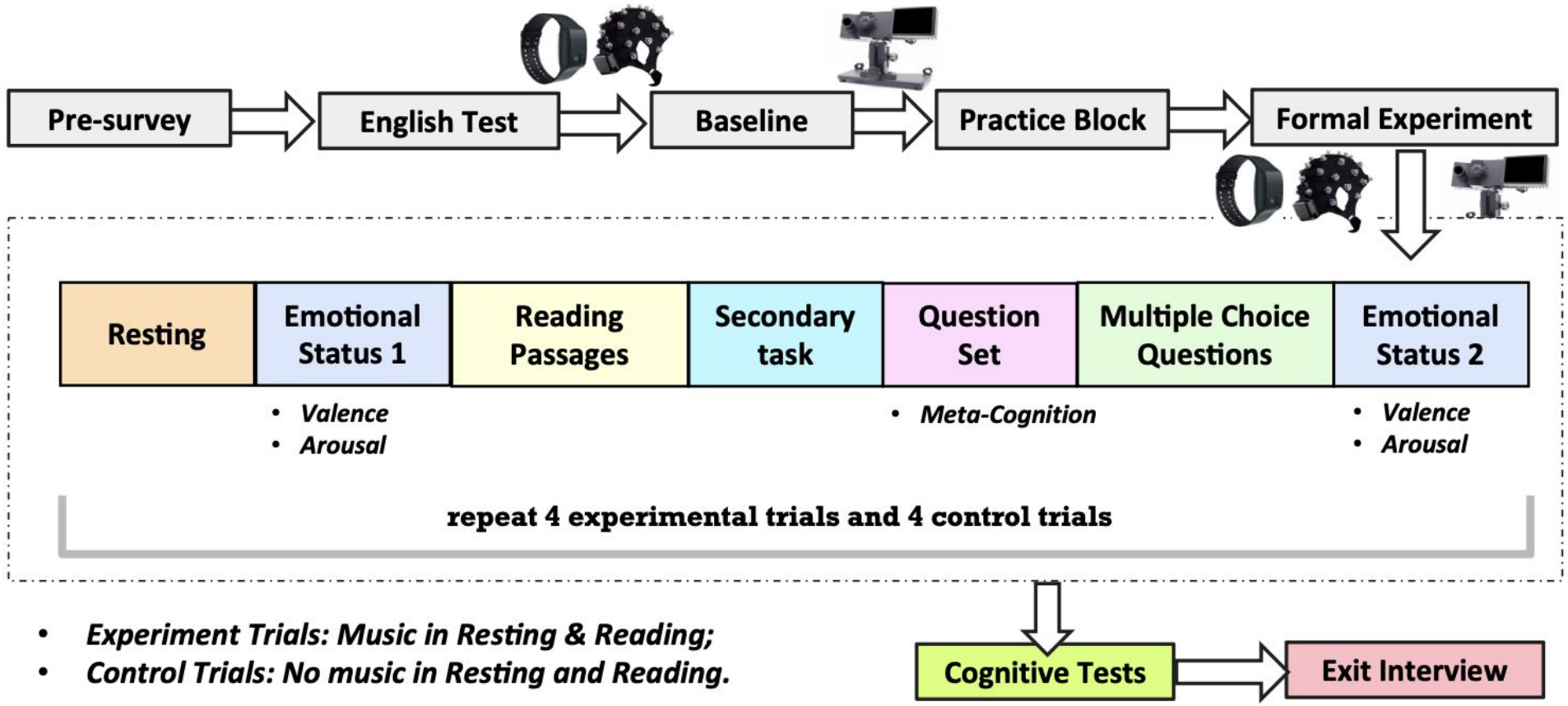
Electroencephalography (EEG) Signals

- Power spectrum analysis
  - Alpha: 9-14 Hz, relaxed, calm
  - Beta: 15-30 Hz, awake, normal alert
  - Gamma: 30 -100 Hz, intensely focused

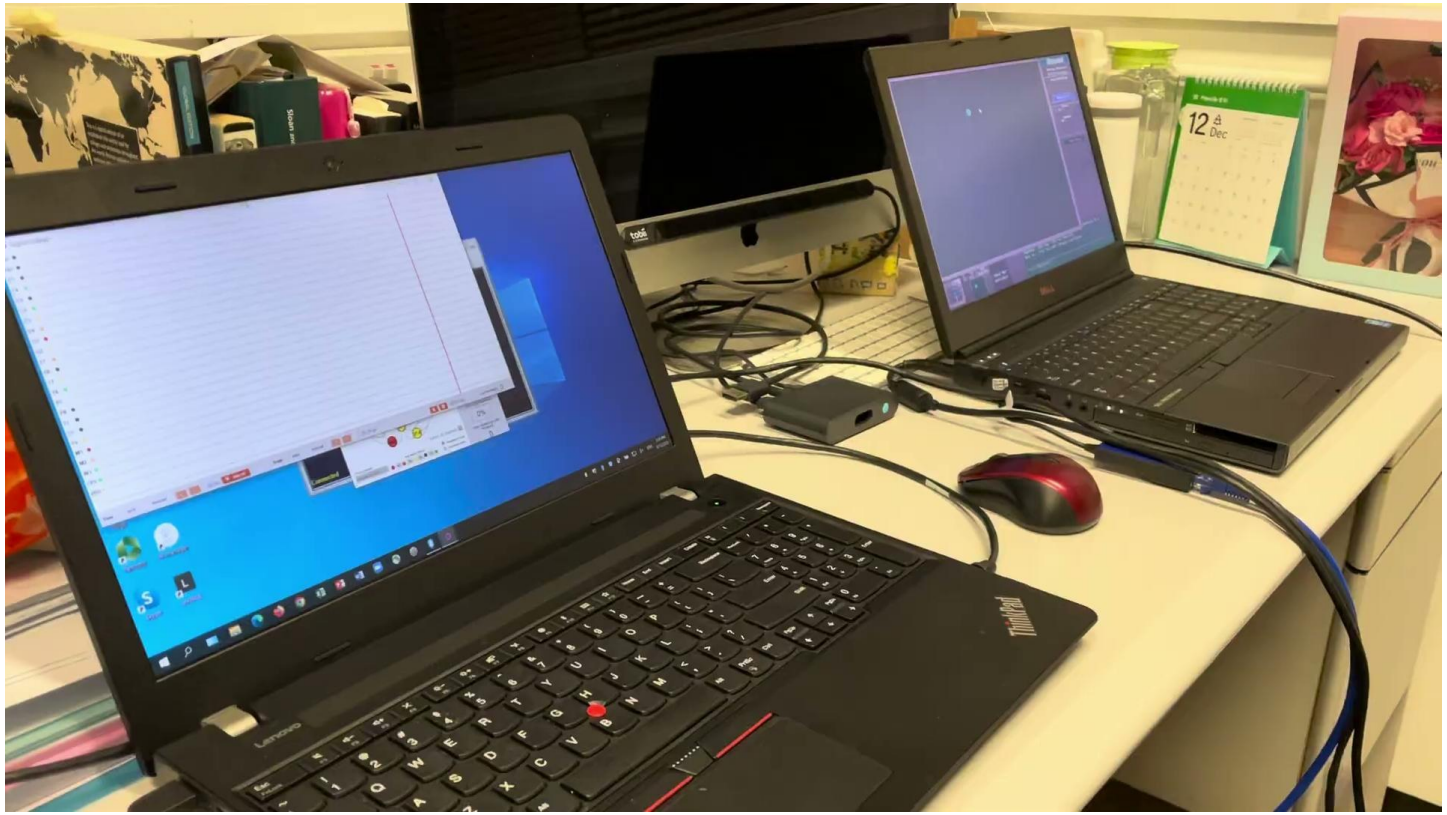


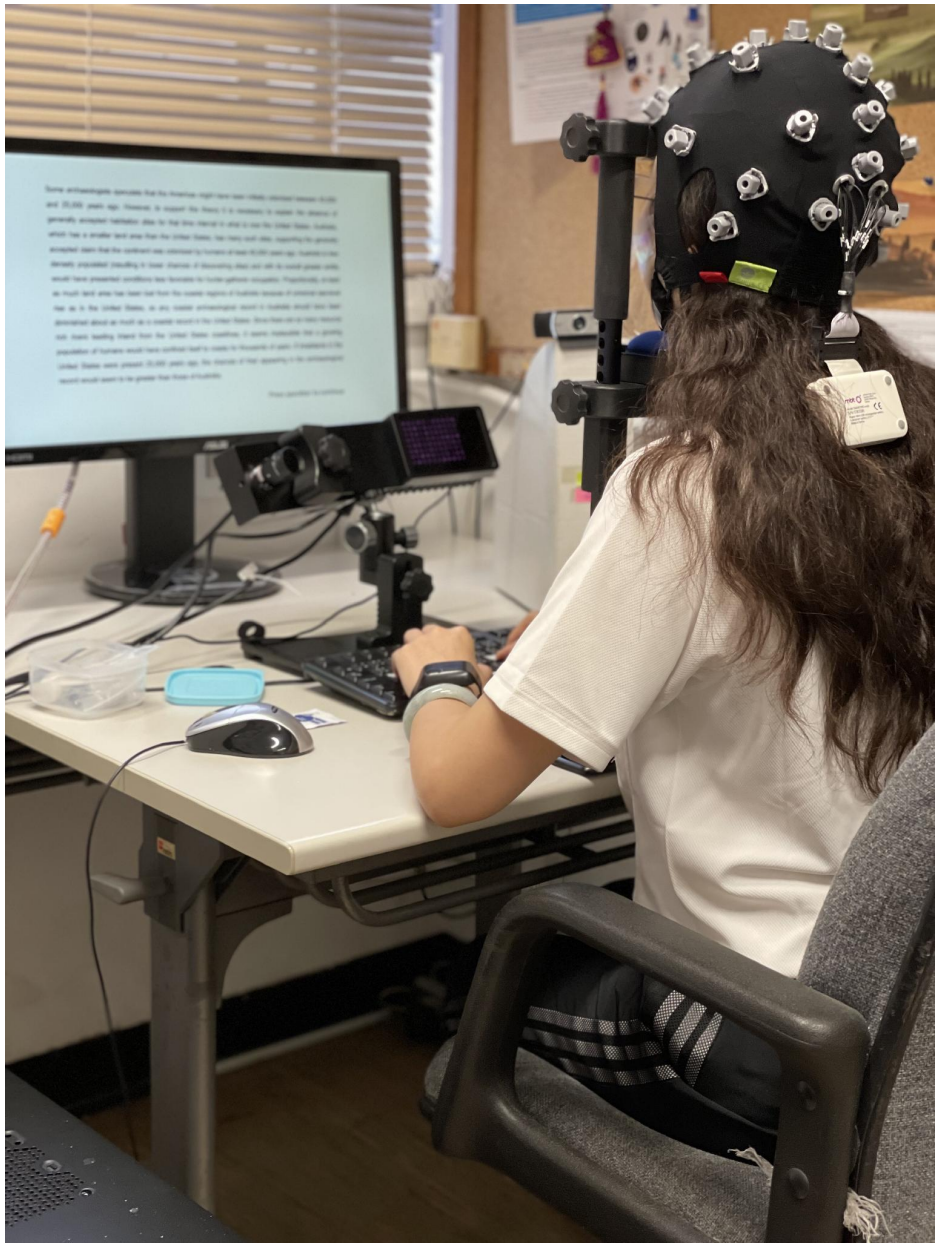
Que, Y., Andrea-Penna, G. M.D., Hu, X., Dong, Y., Chiba, A. A., Iversen, J. R. (2021). Reading with and without background music: An exploration with EEG, eye movement and heart rate, In *Companion Proceedings of the 11th International Conference on Learning Analytics and Knowledge (LAK), CrossMMLA Workshop.*

Study 3-2: Behavioral + Physiological + Eye movement + Brain signals (N = 52)



Experimental setup for multimodal data collection in the music and reading experiment





- Data Analysis
  - Within-subject comparison, music vs. silence.
- Results
  - No **overall** effect on comprehension or reading time
  - No effect on self-reports, eye movements, EEG, heart rate
  - Small effect sizes throughout, true null, not weak power

BGM has No Negative Effect on Reading

Que, Y., & Hu, X. (2025). Enhancing learners' reading comprehension with preferred background music: An eye-tracking, EEG, and heart rate study. *Reading Research Quarterly*, 60(2), e70004.

TABLE 1 | Comparison of results for the music and silence conditions.

Measures	Silence	Music	*t*	*d*	BF_{10}
Performance					
Overall accuracy	0.46 (0.20)	0.46 (0.21)	0.08	0.01	0.15
Text-based accuracy	0.22 (0.13)	0.21 (0.14)	0.32	0.05	0.16
Inference accuracy	0.11 (0.06)	0.11 (0.05)	−0.21	−0.03	0.15
Passage reading time	134.5 (61.1)	133.7 (59.6)	0.23	0.03	0.16
Measures	**Silence**	**Music**	***w***	***r***	**BF_{10}**
Self-report					
Mental effort	3.46 (0.58)	3.61 (0.62)	346	−0.33	0.91
Difficulty	2.81 (0.51)	2.92 (0.65)	305	−0.22	0.40
Understanding	3.37 (0.50)	3.36 (0.54)	481.5	−0.03	0.15
Engagement	3.58 (0.56)	3.72 (0.60)	291	−0.32	1.41
Measures	**Silence**	**Music**	***t***	***d***	**BF_{10}**
Eye movement					
First fixation duration (ms)	242 (28)	242 (29)	0.21	0.03	0.15
First pass dwell time (ms)	288 (40)	285 (41)	1.45	0.20	0.41
First saccade amplitude	3.5 (0.6)	3.6 (0.6)	−1.18	−0.16	0.29
First saccade duration (ms)	51 (44)	46 (22)	1.27	0.18	0.32
Total fixation duration (s)	112 (52)	112 (51)	0.06	0.01	0.15
Total fixation count	469 (199)	470 (205)	−0.10	−0.01	0.15
Total saccade count	468 (199)	469 (205)	−0.10	−0.01	0.15
Total regression count	113 (63)	116 (68)	−0.84	−0.12	0.21
Mean fixation duration (ms)	237 (28)	237 (27)	−0.35	−0.05	0.16
Mean saccade amplitude	3.3 (0.6)	3.3 (0.6)	−0.34	−0.05	0.16
EEG signal					
Fz theta	1.23 (0.65)	1.21 (0.60)	0.74	0.10	0.20
Fz alpha	0.83 (0.86)	0.79 (0.79)	2.22*	0.31	1.42
Fz beta	0.20 (0.12)	0.20 (0.11)	1.77	0.25	0.65
Cz theta	0.69 (0.29)	0.70 (0.30)	−1.44	−0.20	0.40
Cz alpha	0.58 (0.51)	0.56 (0.45)	1.41	0.20	0.38
Cz beta	0.13 (0.07)	0.13 (0.07)	0.81	0.11	0.21
Pz theta	0.66 (0.33)	0.66 (0.33)	0.73	0.10	0.20
Pz alpha	1.04 (2.07)	1.04 (2.11)	0.05	0.01	0.15
Pz beta	0.16 (0.11)	0.16 (0.11)	0.28	0.04	0.16
Heart rates					
Mean HR	76 (10)	76 (11)	−0.62	−0.09	0.18
HRV RMSSD	45 (18)	47 (25)	−0.79	−0.11	0.20

Note: $N = 52$; effect size is given by Cohen's *d* (for paired samples *t*-test) and rank biserial correlation *r* (for Wilcoxon signed-rank test).
*$p < 0.05$.

- Data Analysis
  - Hierarchical linear modeling, learner trait × music condition interactions.

- Results
  - Low working memory / low multitasking: higher cognitive load under BGM
  - High proficiency / rich prior knowledge / BGM habit: more efficient processing
  - Effect direction depends on the learner; positives and negatives cancel out

Que, Y., & Hu, X. (2025). Enhancing learners' reading comprehension with preferred background music: An eye-tracking, EEG, and heart rate study. *Reading Research Quarterly*, 60(2), e70004.

TABLE 3 | BGM condition and individual characteristics interactions in random slope models.

Measures	BGM*2-back RT	BGM*MT RT	BGM*ListenFreq	BGM*LexTale	BGM*PriorKnow
Performance					
Overall accuracy	−0.04	0.14	−0.06	0.05	0.04
Text-based accuracy	−0.10	0.36	−0.04	0.06	0.01
Inference accuracy	0.02	−0.09	−0.07	0.04	0.07
Passage reading time	52.94**	−6.21	−2.95	−74.27**	−1.08
Self-report					
Mental effort	0.51	0.26	−0.09	−0.49	−0.04
Difficulty	0.24	1.38**	−0.05	−0.41	−0.18**
Understanding	0.02	−0.83	−0.01	0.13	0.13**
Engagement	0.35	−1.06**	−0.07	0.33	0.05
Eye movement					
First fixation duration	16.21	8.81	1.49	−37.73**	−0.09
First pass dwell time	33.69**	14.10	3.38	−73.10***	−2.43
First saccade amplitude	−0.32	−0.33	−0.08	1.05***	0.00
First saccade duration	18.01	−15.68	−0.26	−37.81	3.20
Total fixation durations	40275**	−2383	−2191	−52,301	−2399
Total fixation count	112.92	30.56	−9.35	−162.18	−5.25
Total saccade count	113.07	30.62	−9.35	−162.43	−5.25
Total regression count	23.71	−3.47	−6.43	−21.31	2.13
Mean fixation duration	17.81	9.35	0.94	−36.50**	−0.70
Mean saccade amplitude	−0.35	−0.57	−0.08	1.17***	0.01
EEG signal					
Fz theta	−0.15	−0.38	0.09	0.12	0.02
Fz alpha	−0.44	0.74	0.20**	−0.16	−0.06
Fz beta	0.00	−0.14	0.00	0.08	−0.01
Cz theta	0.08	−0.40**	0.04	−0.00	0.00
Cz alpha	−0.34**	0.42	0.09**	0.13	−0.03
Cz beta	0.05	−0.08	−0.00	−0.01	−0.00
Pz theta	−0.09	−0.41**	0.02	0.40**	−0.02
Pz alpha	−1.00	1.08	0.56***	0.16	−0.22
Pz beta	0.05	−0.17**	−0.01	0.08	−0.01
Heart rates					
Mean HR	4.22	−4.34	1.57	−9.57	0.14
HRV RMSSD	−26.10***	0.17	−0.66	53.71***	−0.96

Note: $N = 52$.

Abbreviations: 2-back RT = 2-back test response time, LexTale = English proficiency measured by LexTale score, ListenFreq = BGM listening frequency, MT RT = multitasking test response time, PriorKnow = prior knowledge.

**$p < 0.01$.

***$p < 0.001$.

- Data Analysis
  - Stepwise regression on acoustic features

- Results
  - Lyrics: lower accuracy, slower lexical processing, higher mental demand
  - Lyrics: fewer but longer fixations
  - Fast tempo: broader saccades, shorter reading time (more efficient)

TABLE 4 | Linear regression models.

Modality	Targets	Predictors	*B*	Error	Beta	*t*	Sig.	R^2
Performance	Overall Acc	Instrumentation	−0.11	0.05	−0.14	−2.10	0.037	0.02
Eye movement	First FixDur	Instrumentation	16.62	4.03	0.28	4.12	<0.001	0.08
	First dwell time	Instrumentation	19.16	5.89	0.22	3.26	0.001	0.09
		Tempo	−0.23	0.09	−0.17	−2.56	0.010	
	First SacAmp	Tempo	0.00	0.00	0.17	2.47	0.014	0.03
	Mean FixDur	Instrumentation	13.94	3.78	0.25	3.69	<0.001	0.06
	Mean SacAmp	Tempo	0.00	0.00	0.18	2.67	0.008	0.03
	Log_{10}TotFixDur	Tempo	−0.00	0.00	−0.14	−2.01	0.045	0.02
	Log_{10}TotFixCnt	Instrumentation	−0.08	0.03	−0.18	−2.59	0.010	0.05
		Tempo	−0.00	0.00	−0.15	−2.12	0.035	
	Log_{10}TotSacCnt	Instrumentation	−0.08	0.03	−0.18	−2.59	0.010	0.05
		Tempo	−0.00	0.00	−0.15	−2.12	0.035	
	Log_{10}TotRegCnt	Tempo	−0.07	0.04	−0.14	−1.99	0.048	0.02
Heart rate	Mean HR	Instrumentation	3.72	1.56	0.16	2.38	0.018	0.03

Abbreviations: Acc = accuracy, B = unstandardized B, Beta = standardized coefficient beta, Error = coefficient standard error, FixDur = fixation duration, SacAmp = saccade amplitude.

Lyrics hurt, Fast tempo helped

Que, Y., & Hu, X. (2025). Enhancing learners' reading comprehension with preferred background music: An eye-tracking, EEG, and heart rate study. *Reading Research Quarterly*, 60(2), e70004.

- Data Analysis
  - Thematic analysis of user interviews

- Results
  - Half report benefits: concentration, efficiency, noise blocking
  - Two-thirds report harm: distraction, attributed to lyrics, fast tempo, loudness
  - Small group report no impact or longer time

TABLE 5 | Interview coding results on effects of BGM on learners' cognition.

Code	Definition	Parti CNT	Excpt CNT
Category: Benefits		25	37
Concentration	Participant mentioned BGM helped increase concentration during reading	17	19
Efficiency	Participant mentioned BGM improved reading efficiency	8	8
Block out noise	Participant mentioned BGM helped block out noise when they were reading the text	5	7
Interestingness	Participant mentioned BGM increased interestingness of text reading	2	3
Category: Harm		36	51
Distraction	Participant mentioned BGM distracted reading	33	47
Increase cognitive load	Participant mentioned BGM increased cognitive load during reading comprehension	3	3
Impede word recognition	Participant mentioned BGM impeded word recognition	1	1
Category: Neutral		12	13
No impact	Participant mentioned BGM did not affect their comprehension extent, cognitive load, and cognitive processes	11	11
Read for longer	Participant mentioned BGM led to a longer reading time	2	2

Abbreviations: Excpt CNT = excerpt count (total = 101), Parti CNT = participant count (total = 52).

Que, Y., & Hu, X. (2025). Enhancing learners' reading comprehension with preferred background music: An eye-tracking, EEG, and heart rate study. *Reading Research Quarterly*, 60(2), e70004.

Music for Learning in Virtual Reality

Immersion



© Medium

Interaction



© Creative Applications

Imagination



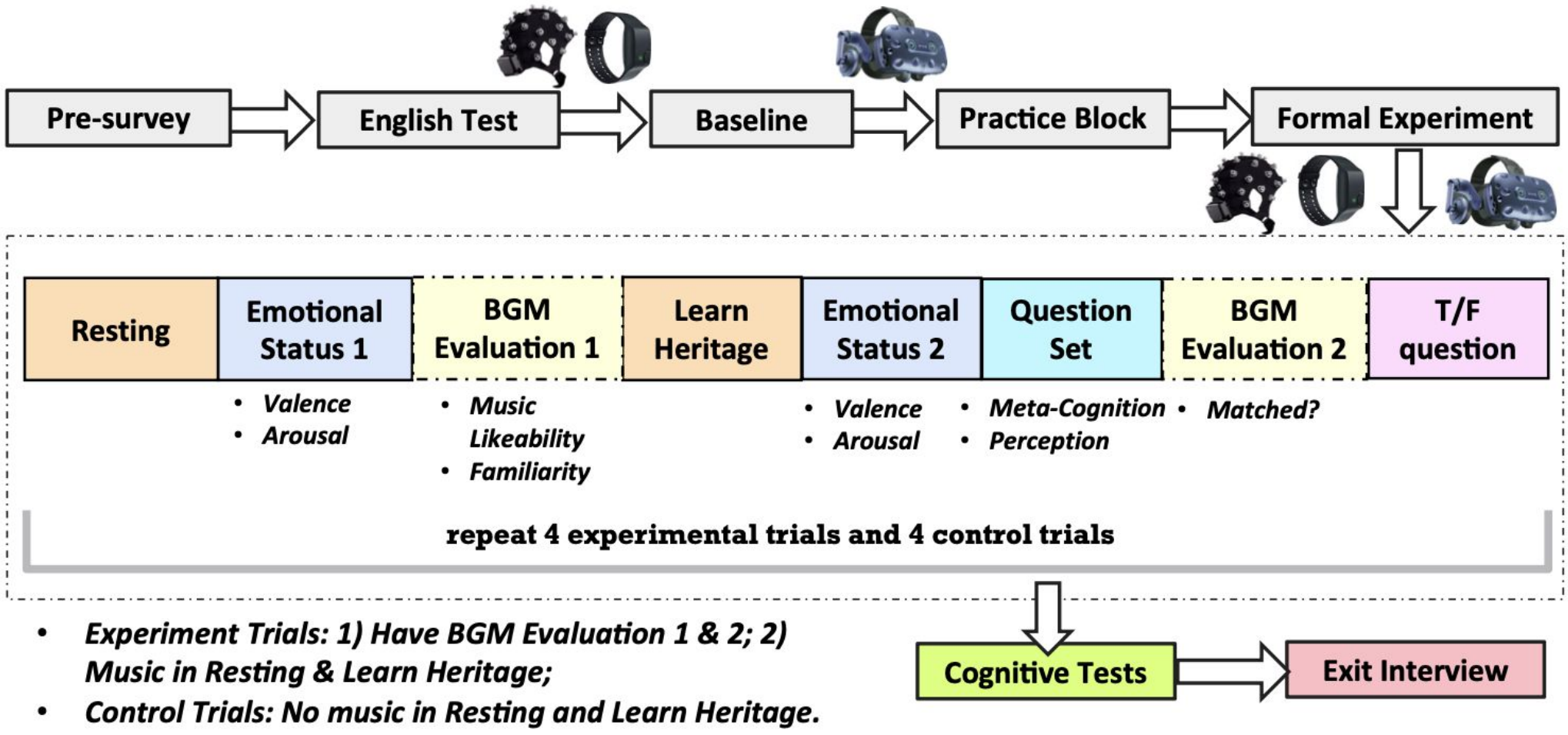
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Study 4: Behavioral + Physiological + Eye movement + Brain signals in VR (N = 52)

- BGM for VR heritage appreciation
- Physiological + Eye movement + EEG



Experiment procedure for BGM in VR



- ***Experiment Trials: 1) Have BGM Evaluation 1 & 2; 2) Music in Resting & Learn Heritage;***
- ***Control Trials: No music in Resting and Learn Heritage.***

- BGM vs. Silence
  - No difference in comprehension accuracy or viewing time
  - Higher enjoyment under BGM
  - Higher perceived immersiveness under BGM
  - No change in valence or arousal

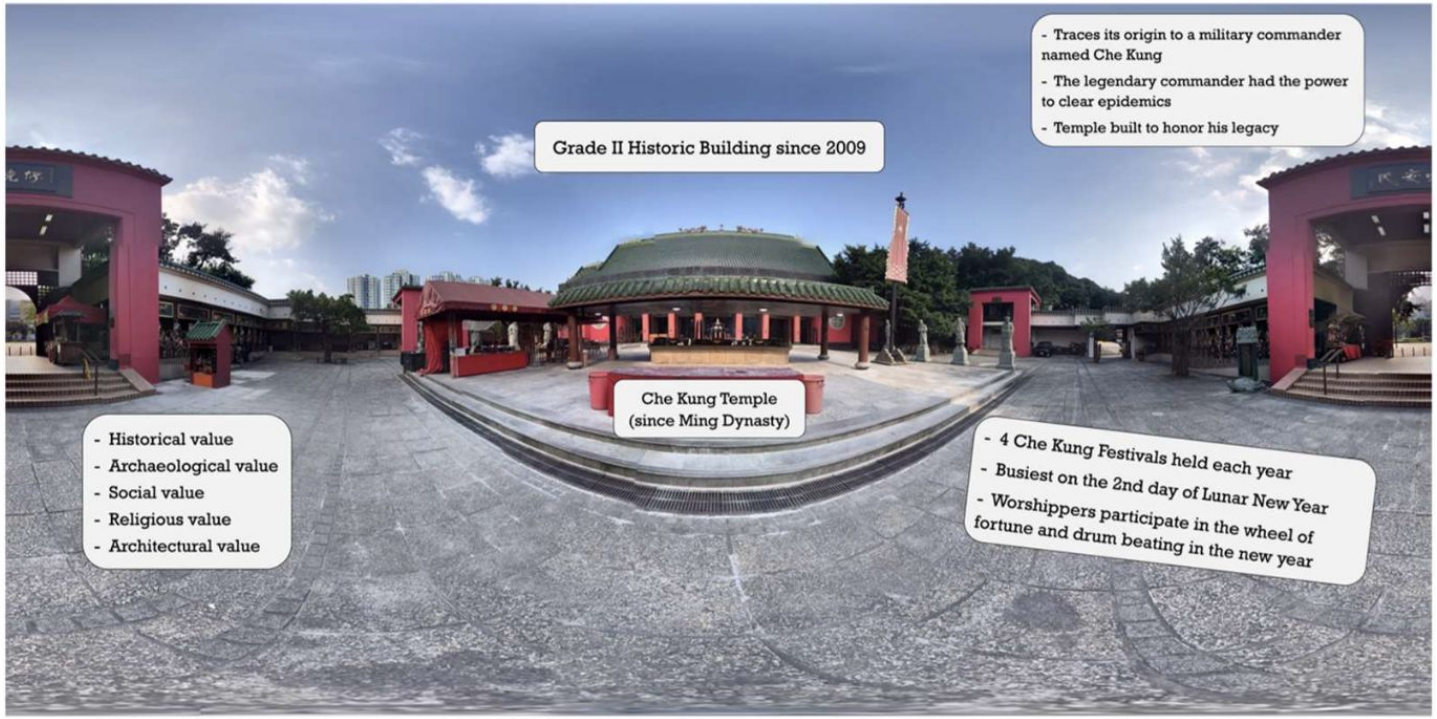


Figure 1. Sample VR Scene with Text Annotations.

Que, Y., & Hu, X. (2025). Exploring background music for studying in virtual reality learning environments. *Interactive Learning Environments*, 33(9), 5516-5529.

Data Analysis & Results

- Music characteristics and learning outcomes
  - Repeated-measures correlation (rmcorr)
  - For intra-individual associations
  - Benjamini–Hochberg FDR = .05
- Results
  - BGM likeability: higher text-based accuracy
  - Faster tempo: lower immersion and enjoyment
  - Greater BGM familiarity: smaller arousal increase
  - Perceived scene–music compatibility: no relationship with any outcome

Que, Y., & Hu, X. (2025). Exploring background music for studying in virtual reality learning environments. Interactive Learning Environments, 33(9), 5516-5529.

Data Analysis & Results

- Predict learning outcomes using learner traits
  - Stepwise linear regression
  - Data from the BGM condition only
- Results
  - Higher BGM listening frequency: higher accuracy, immersion, enjoyment
  - Higher English proficiency: shorter viewing time

Que, Y., & Hu, X. (2025). Exploring background music for studying in virtual reality learning environments. Interactive Learning Environments, 33(9), 5516-5529.

Data Analysis & Results

- **Eye movements:** Comparison for text and image area of interest (AOIs)
- Results
  - BGM increased mean fixation duration on images
    - BGM may have facilitated deeper visual processing of images in virtual reality.
  - No significant effects on text-AOI measures.

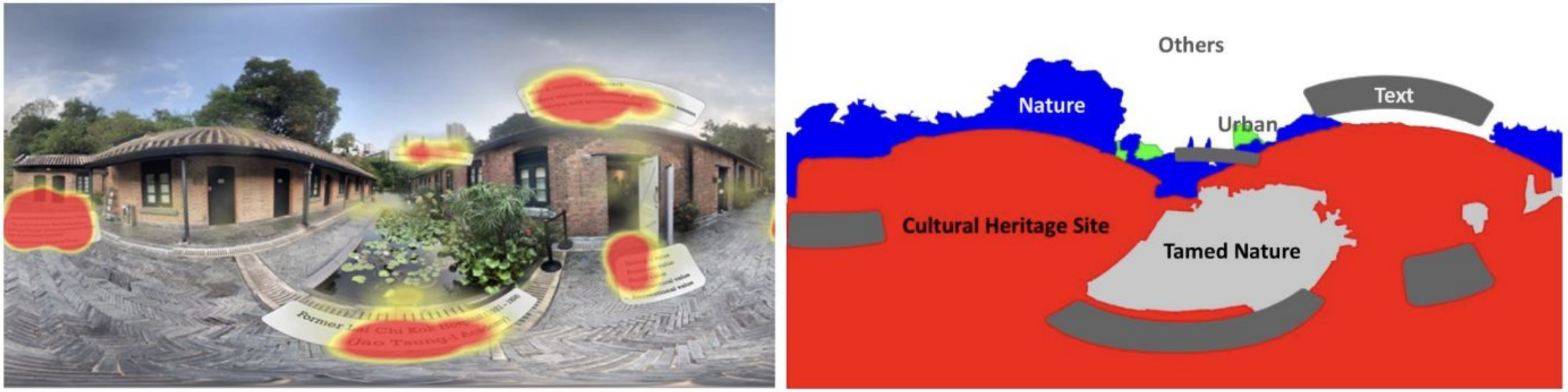


Figure 5. Sample VR scene and AOIs adapted from Ng et al. (2022).

Que, Y., & Hu, X. (2026). Examining the effect of background music on learners' attention and cognition in virtual reality environments: A psychophysiological study. *International Journal of Human–Computer Interaction*, 42(1), 211-224.

Data Analysis & Results

- EEG signals: EEGLAB preprocessing; Extracted mean log spectral power in theta, alpha, and beta bands at Fz, Cz, and Pz.
- Result:
  - No significant difference between BGM and silence conditions
  - Habitual BGM listening negatively moderated BGM effects on Fz alpha and Fz/Cz beta power.
- Meaning of the indicator:
  - Lower alpha and beta power may indicate a brain state conducive to information processing and memory encoding.
- Interpretation:
  - Habitual listeners may adapt their neural processing to learning with BGM.

Que, Y., & Hu, X. (2026). Examining the effect of background music on learners' attention and cognition in virtual reality environments: A psychophysiological study. *International Journal of Human–Computer Interaction*, 42(1), 211-224.

Data Analysis & Results

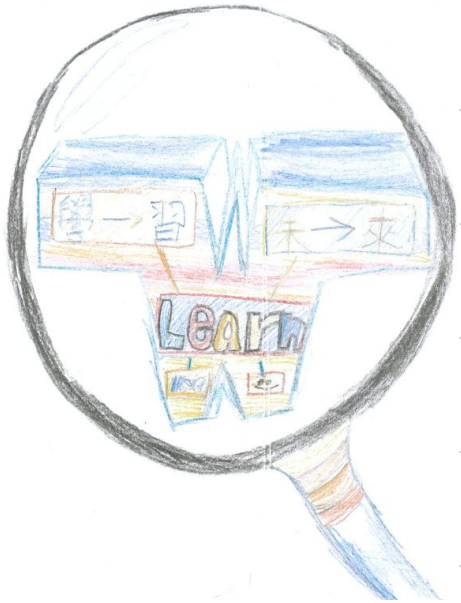
- HR and HRV signals: Screened IBI artefacts and corrected abnormal intervals using linear interpolation; Extract mean HR and HRV RMSSD for each VR scene.
- Result:
  - No significant overall BGM effect on HR or HRV.
  - Habitual BGM listening positively moderated the BGM–HRV RMSSD relationship
- Meaning of the indicator:
  - Increased HRV, indicating greater parasympathetic nervous system activity, associated with the relaxation and restorative process
- Interpretation:
  - BGM may support relaxation and stress regulation, particularly for habitual BGM listeners.

Que, Y., & Hu, X. (2026). Examining the effect of background music on learners' attention and cognition in virtual reality environments: A psychophysiological study. *International Journal of Human–Computer Interaction*, 42(1), 211-224.

Take-aways from Lab Experiments

- Careful design to isolate factors
  - Tasks
  - Music
  - Data collection protocol and device
  - Data analysis
  - …
- When carrying out the experiment
  - Rest state physiological signals (for normalization)
  - Eye tracker calibration and re-calibrate
  - Practice block

Agenda



Time	Session	Presenter
14:00–14:10	1 Welcome & Self-introduction	Xiao Hu
14:10–14:40	2 Foundations	Xiao Hu
14:40–15:20	3 Laboratory experiments	Ying Que
15:20–16:00	4 Naturalistic studies	Fanjie Li & Xiao Hu
16:00–16:15	Coffee break	—
16:15–16:45	5 User-centered measurement	Ying Que
16:45–17:15	6 Hands-on demo	Ying Que & Fanjie Li
17:15–17:30	7 Closing discussion	Xiao Hu, Ying Que, Fanjie Li

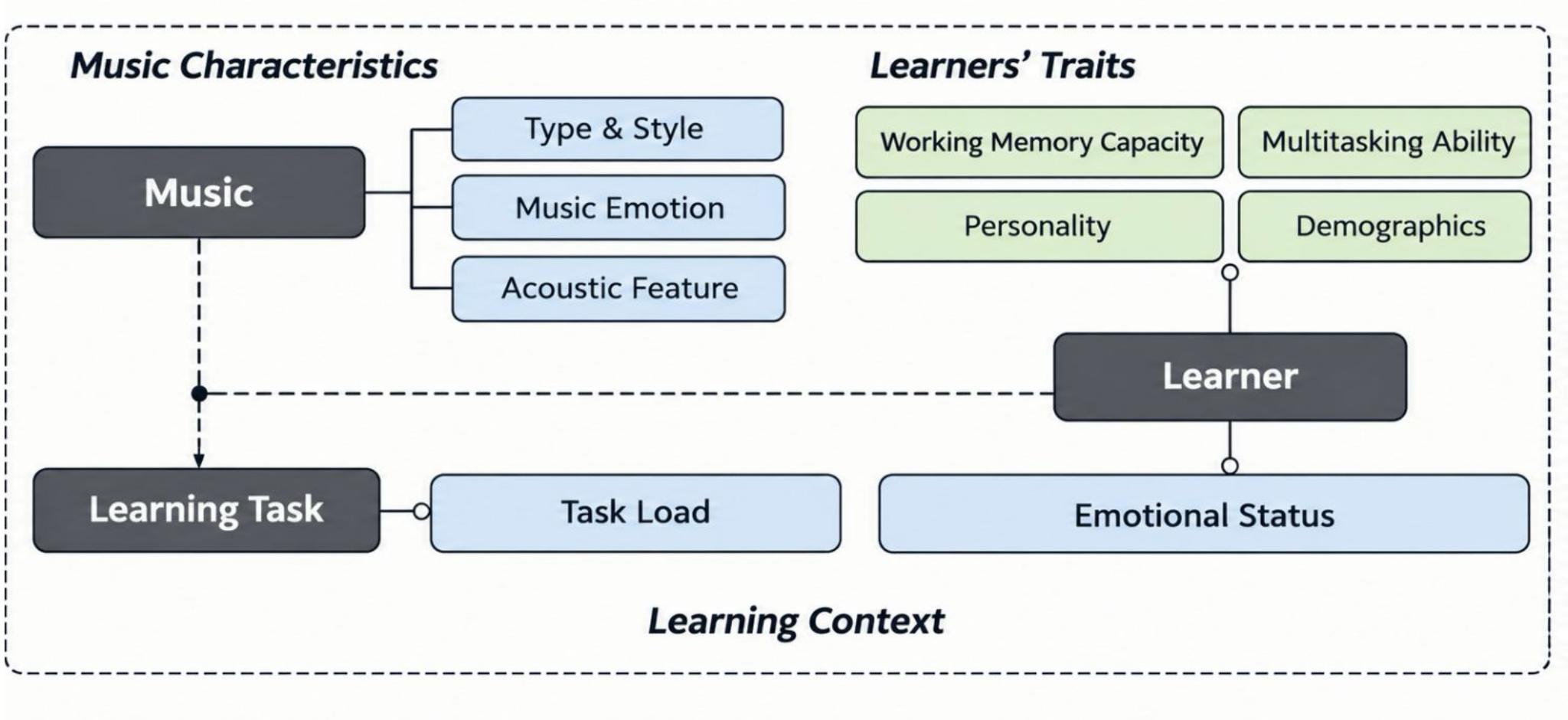
Pros and Cons of Naturalistic Experiment

- Higher external/ecological validity (real-world generalizability)
- Lower participation bias (in their own environments)

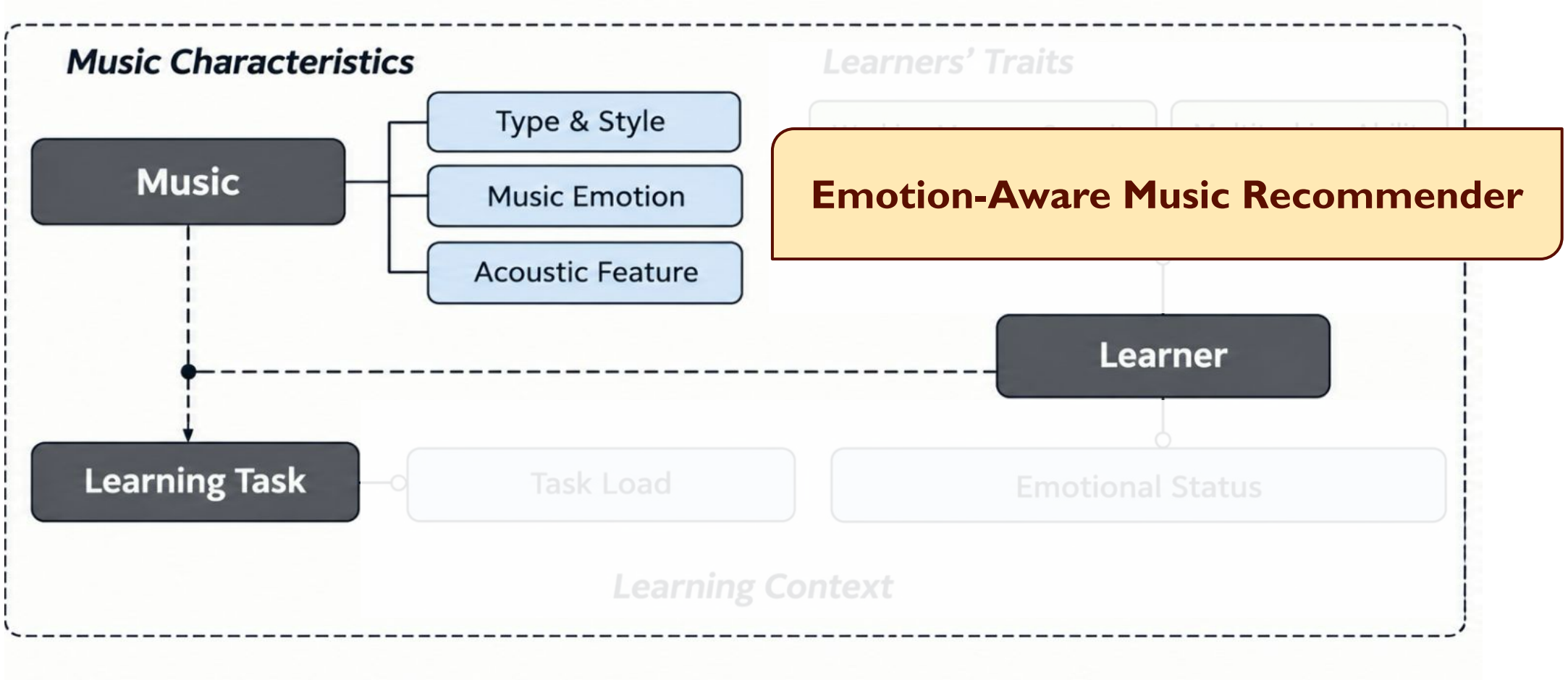
- Lower control of experiment environments
- Lower control of confounding variables
- Constraints of mobile / wireless consumer-grade devices

Therefore, naturalistic experiments often have simpler tasks, and rely on mobile apps

Music, Users, and Tasks

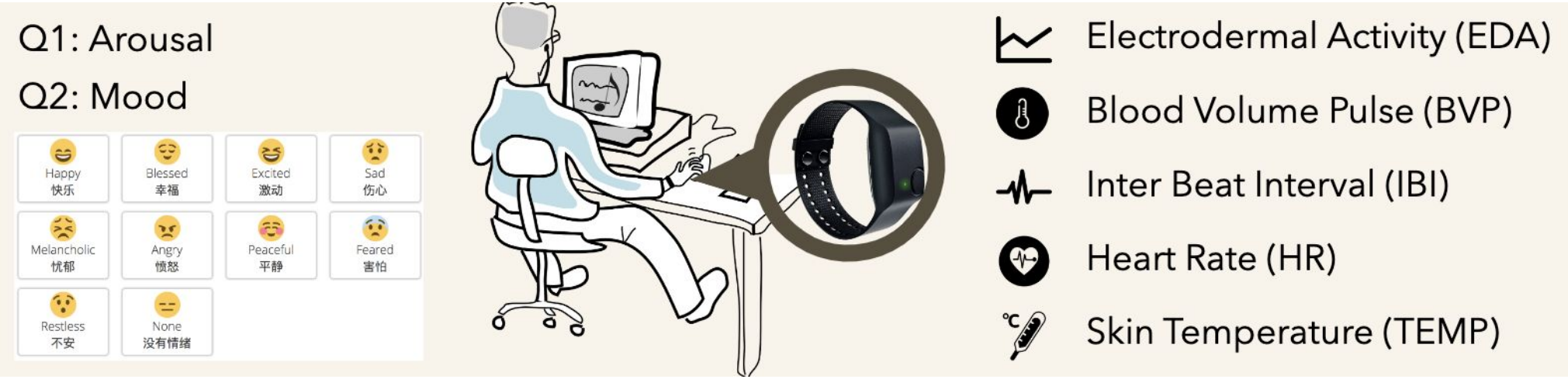


Challenge 1: Support Music Discovery



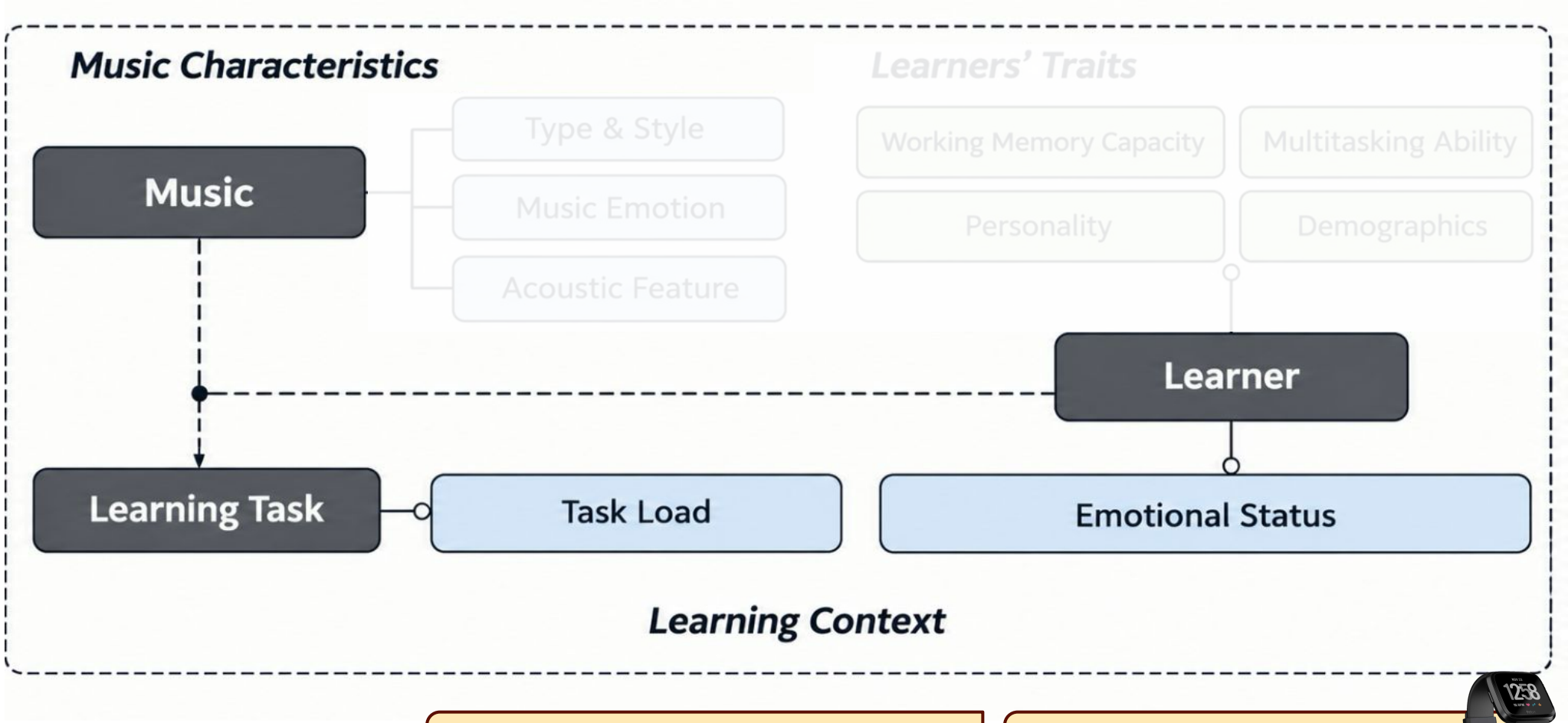
Study 5: Music Recommendations for Enhancing Learning and Well-being

- Connections between music-induced emotion and physiological signals



Hu, X., Li, F., & Ng, J. (2018). On the relationships between music-induced emotion and physiological signals. In *Proceedings of the 19th International Society for Music Information Retrieval Conference (ISMIR)*, Sept. 23 – 27.

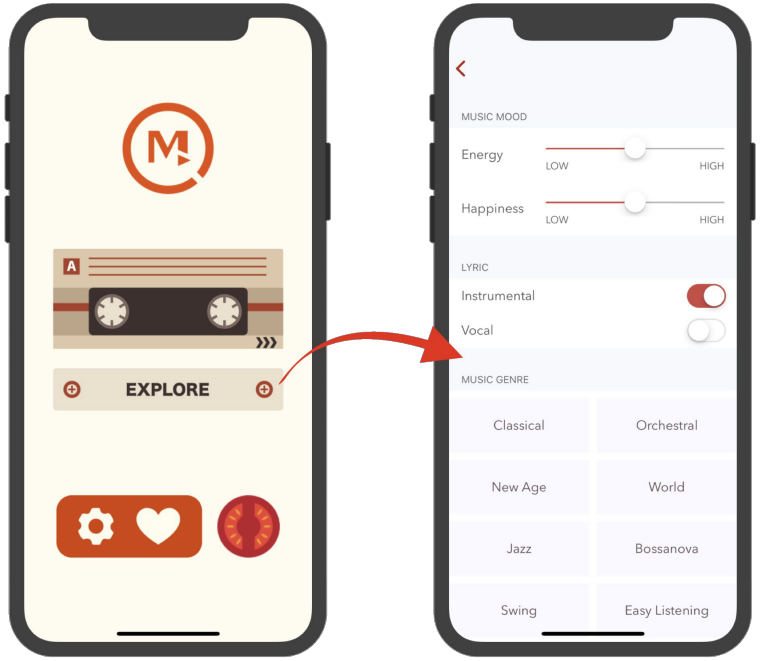
Challenge 2: Capture User Context

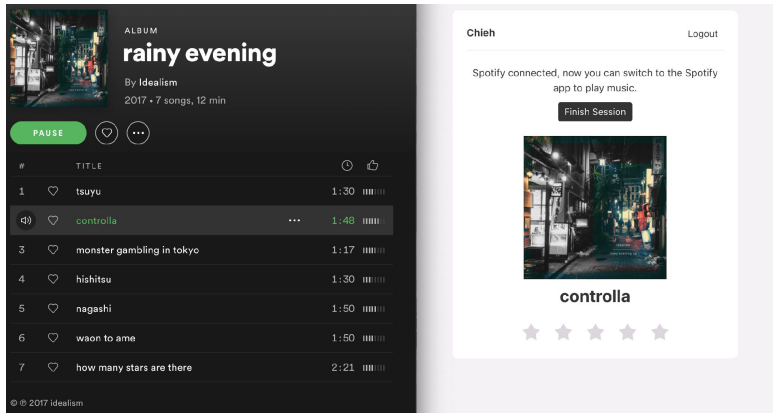


Pop-up Surveys

Fitbit Wristband

Study 6: User-Music Interactions in **Naturalistic** Settings









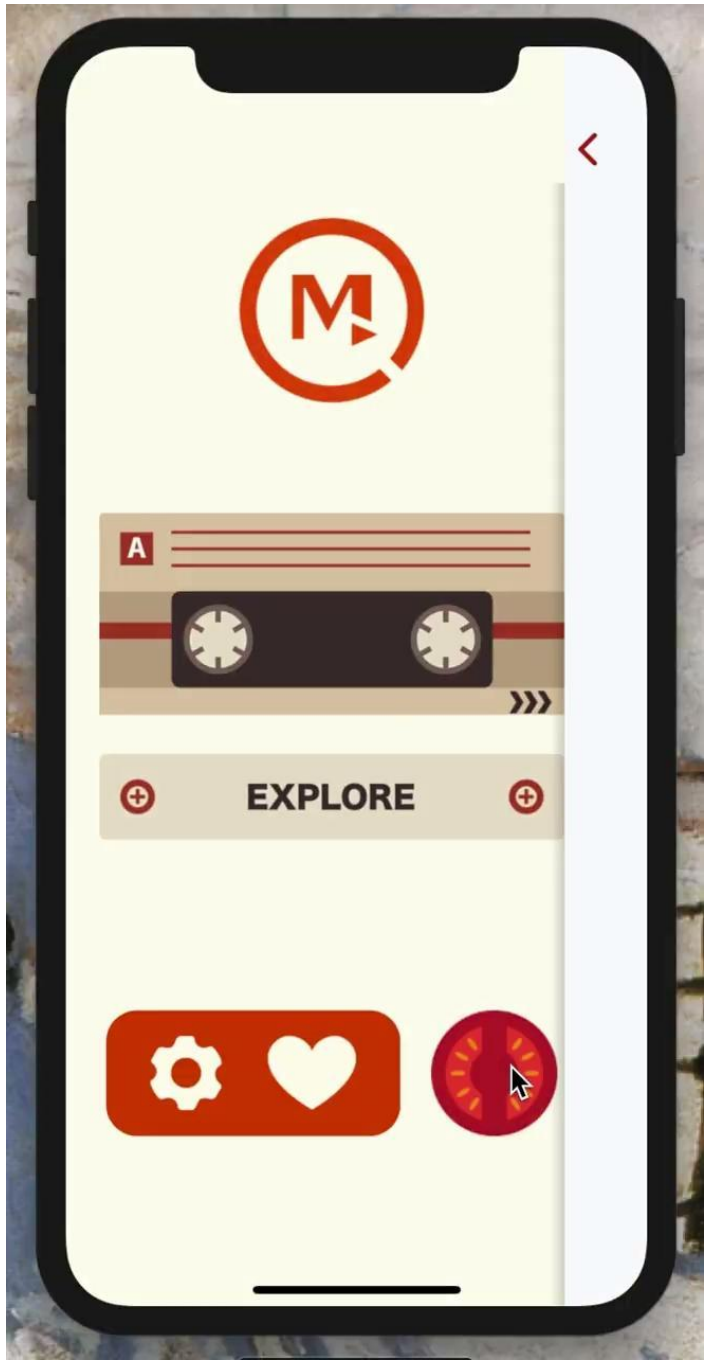


Li, F., Hu, X., & Que, Y. (2020). Learning with background music: A field experiment. In *Proceedings of the 10th International Conference on Learning Analytics & Knowledge (LAK '20)* (pp.224-229) ACM.

Li, F., Wang, Z., Ng, J. T. D. & Hu, X. (2021). Studying with learners' own music: Preliminary findings on concentration and task load, *LAK'21*. (pp.613-229) ACM.

Moody app demo

Li, F., & Hu, X. (2023). Background music for studying: A naturalistic experiment on music characteristics and user perception. *IEEE Multimedia*, 30(1).



N = 30 students

One week: 195 sessions

2618 (1525 unique) music tracks played



TABLE IV

CORRELATIONS BETWEEN MUSIC FEATURES AND LEARNING EXPERIENCE UNDER LOW TASK LOAD

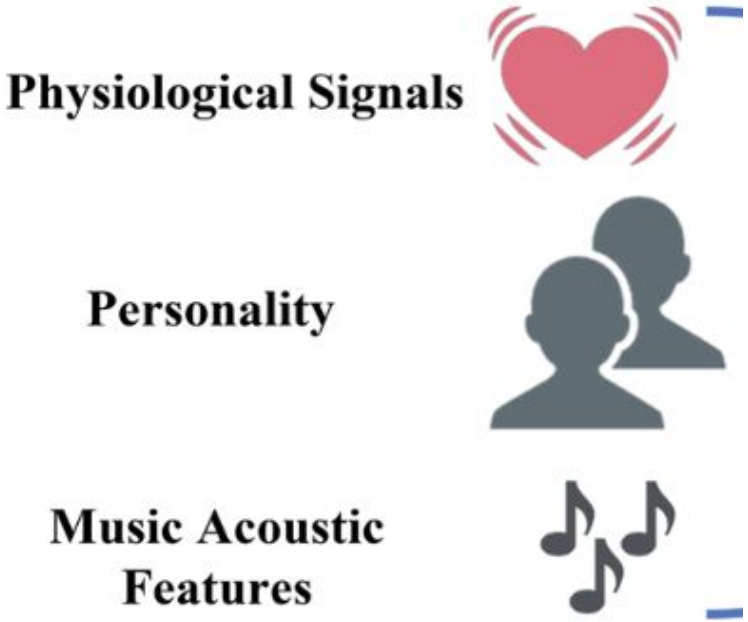
Music features	Low mental demand			Low temporal demand		
	PLE	AST	CON	PLE	AST	CON
Dynamics						
rmse_std	-.129	.111	.094	.060	.207	.231*
Rhythm						
tempo	-.056	-.061	.004	-.130	-.015	.051
avg_onset_frq	-.078	-.071	-.092	-.045	-.120	-.065
Timbre						
rolloff_mean	-.033	.190	.275*	.051	.299*	.285*
rolloff_std	-.200	-.018	.017	-.022	.005	-.037
centroid_mean	-.037	.183	.268*	.051	.302**	.297*
centroid_std	-.207	.043	.108	-.057	.082	.041
flatness_mean	-.051	.154	.289**	.080	.241	.304**
flatness_std	-.092	-.005	.072	-.051	-.003	.019
zcr_mean	-.032	.138	.201	.056	.285*	.278*
zcr_std	-.152	.091	.123	-.054	.141	.096

Note: N=30, L=111 (low mental demand), L=100 (low temporal demand); * $p < .05$, ** $p < .01$.

- Music had no effect on high demand tasks
- Under low demand tasks
  - brighter music enhanced tasks engagement

Li, F., & Hu, X. (2023). Background music for studying: A naturalistic experiment on music characteristics and user perception. *IEEE Multimedia*, 30(1).

Study 7: Incorporating User Context Information in Recommenders



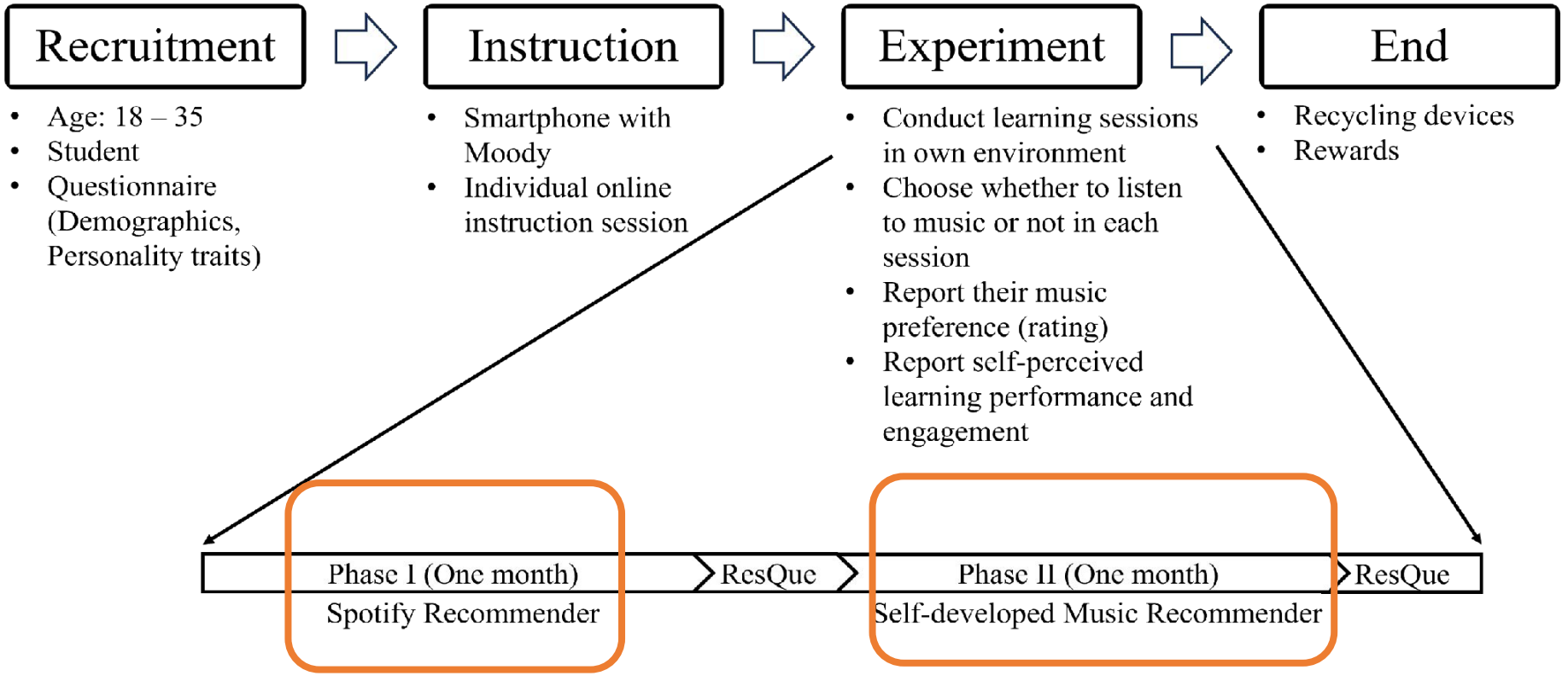
In Predicting

importance
0.306
0.224
0.178

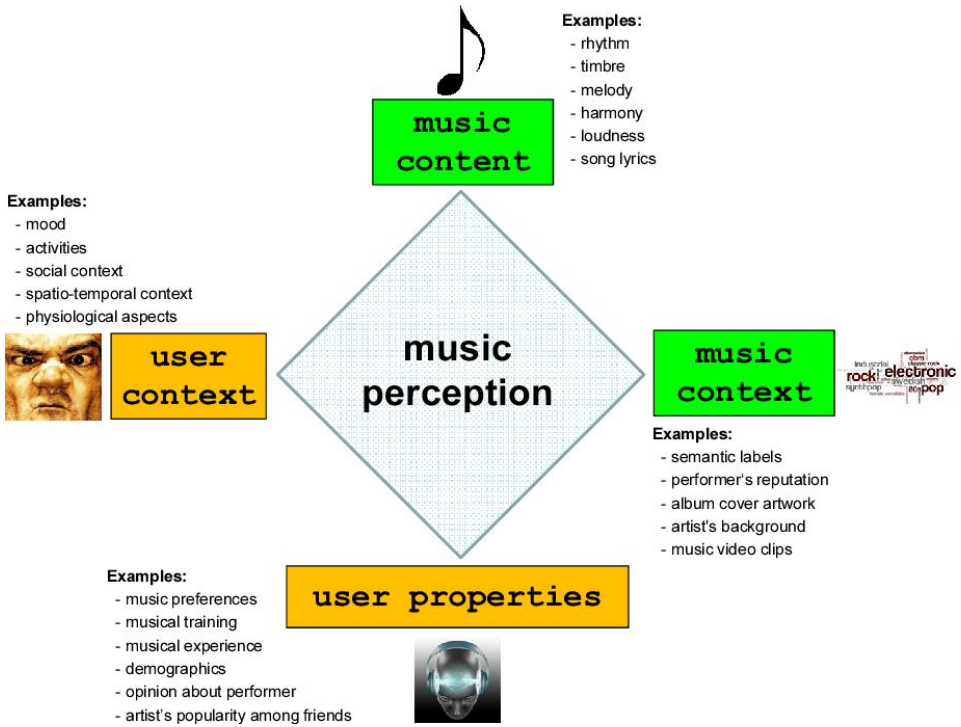
User behavior
Added music to playlist
Played the music from start to end
Played the music for 30 seconds or longer before skipping
Played the music for 10 to 30 seconds (exclusive)
Played the music for shorter than 10 seconds

Liu, R. & Hu, X. (2020). A multimodal music recommendation system with listeners' personality and physiological signals, In *Proceedings of the ACM/IEEE Joint Conference on Digital Libraries (JCDL).*

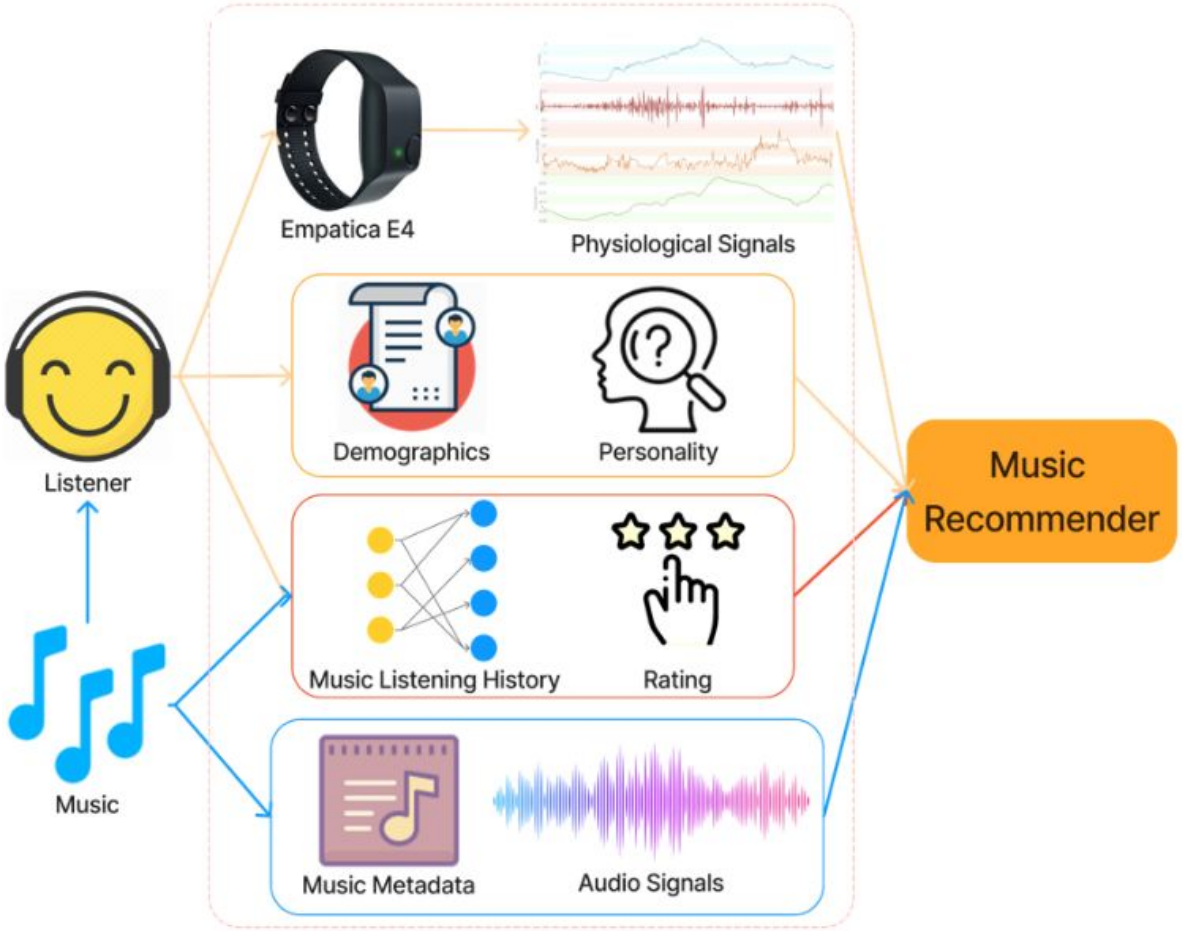
Study 8: A Two-stage Field Experiment on Multimodal Music Recommendation



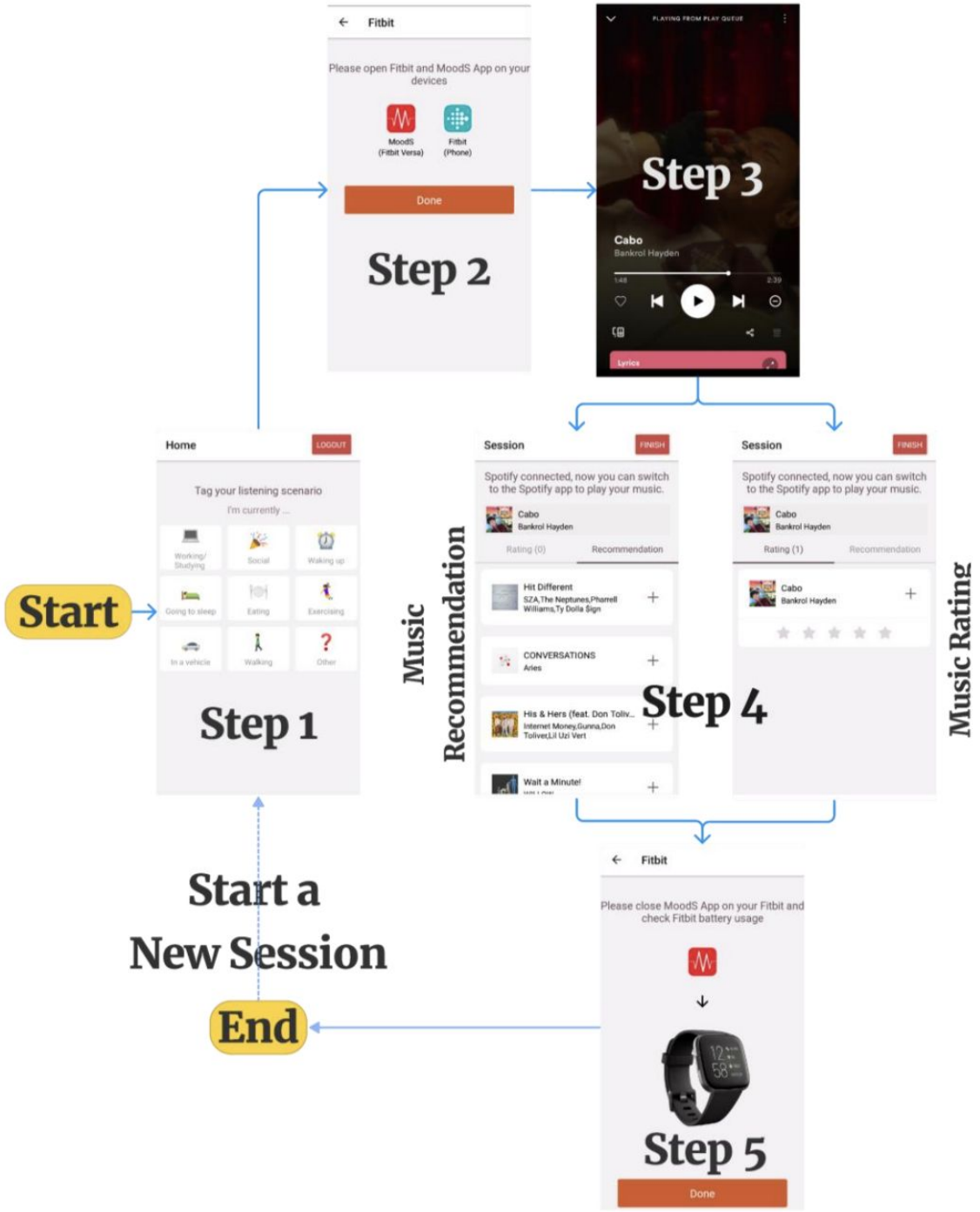
Multimodal Music Recommendation



Human music perception model
(Schedl, Flexer & Urbano, 2013)

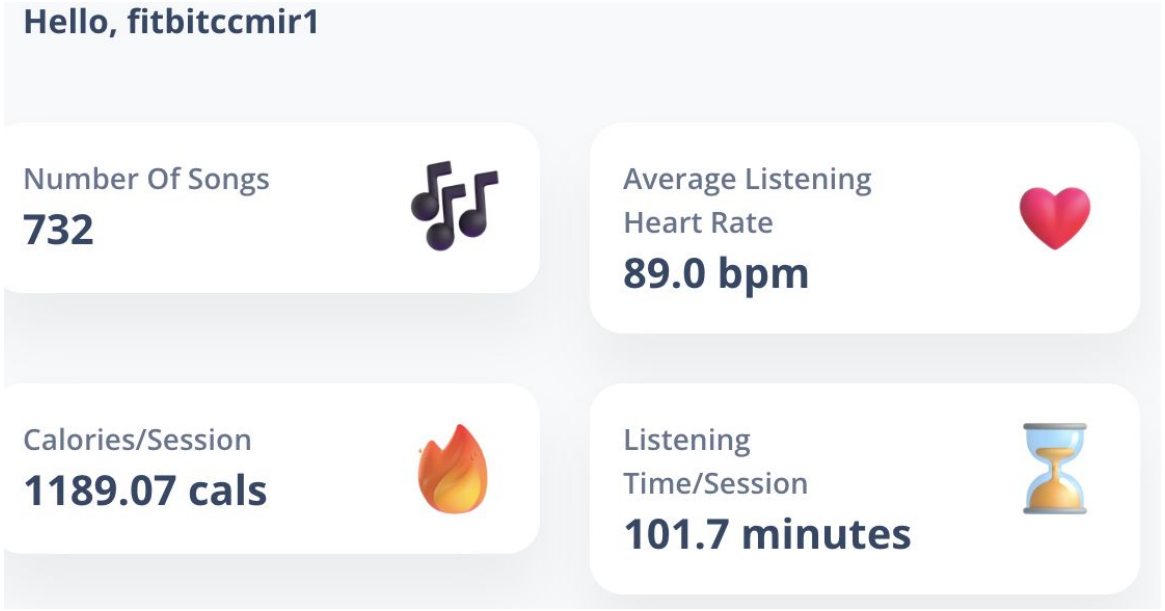


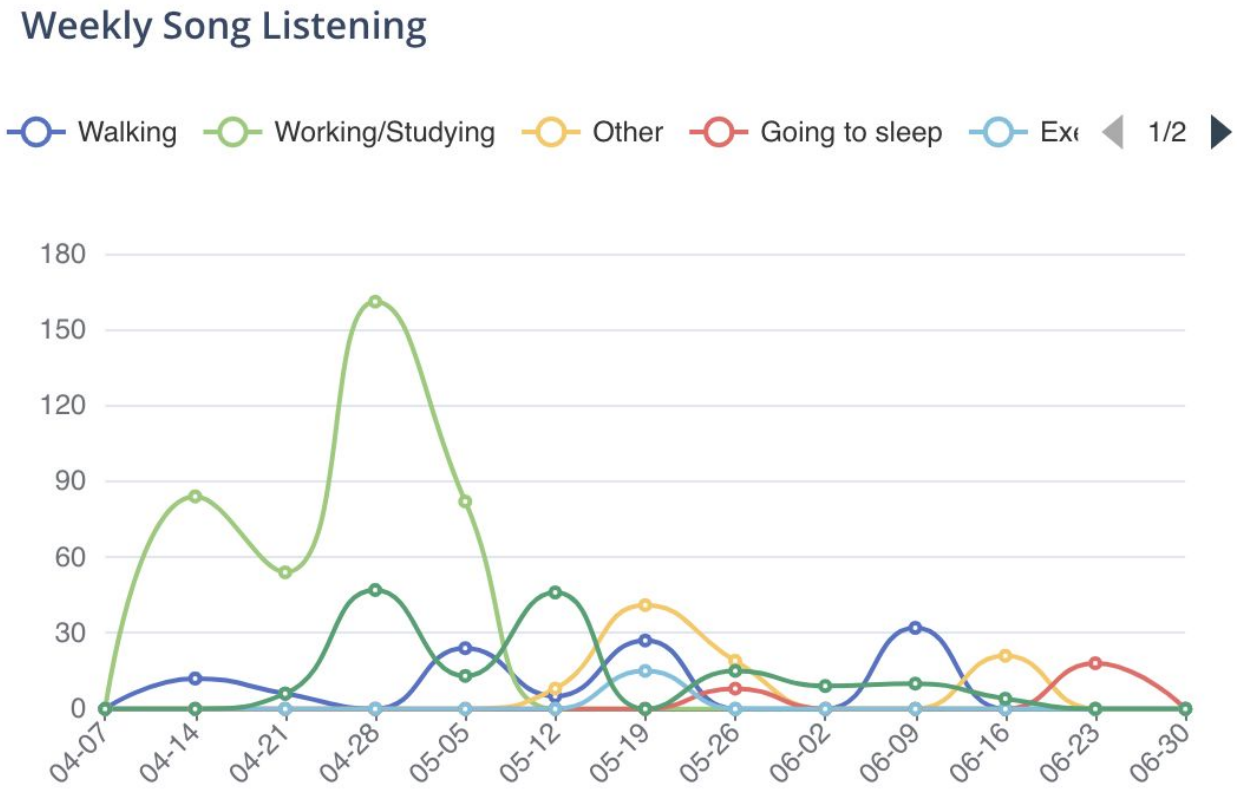
Moody App 2.0



Liu, R., Wang, Y. & Hu, X. (In preparation). A multimodal music recommendation system with physiological signals.

Multimodal Data and Listening Scenarios

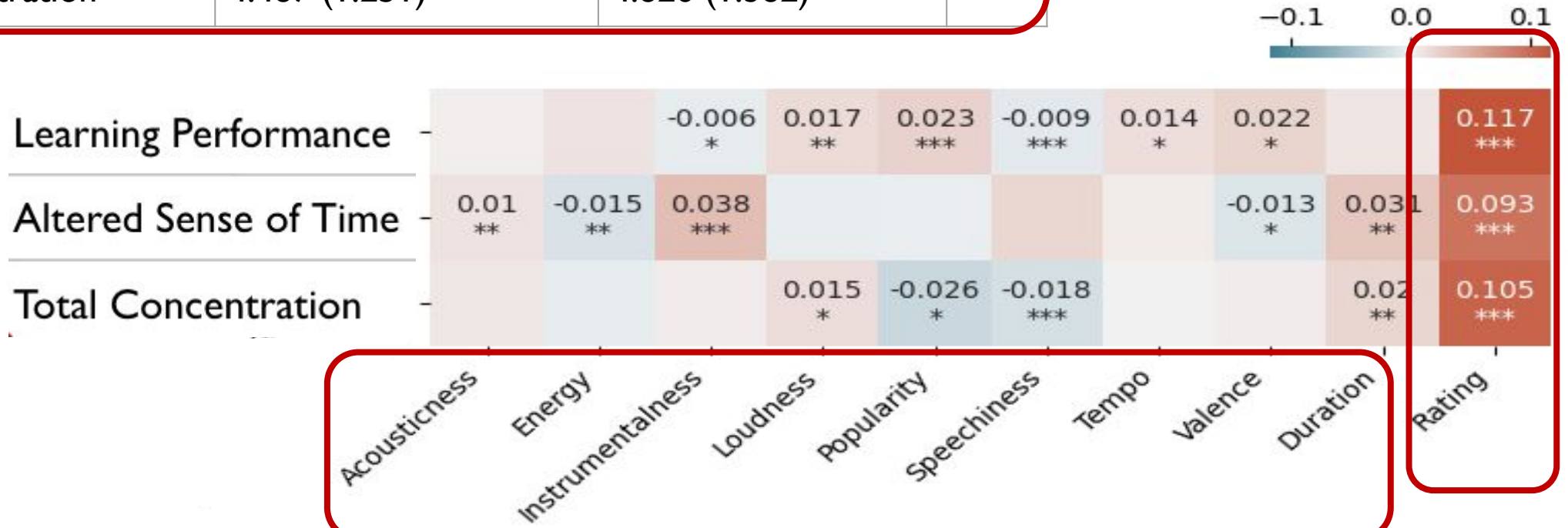




Comparisons between BGM and Silence on Learning

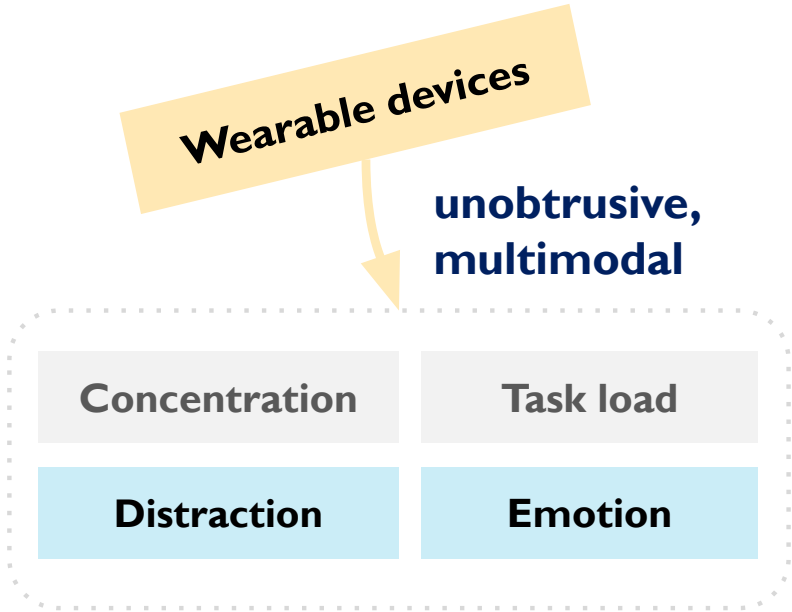
Learning Performance & Engagement	Learning without Music (N = 158) Mean (SD)	Learning with Music (N = 671) Mean (SD)	Sig.
Learning Performance	2.987 (1.612)	4.751 (1.550)	***
Altered Sense of Time	4.162 (1.241)	4.511 (1.339)	***
Total Concentration	4.487 (1.251)	4.826 (1.382)	**

Liu, R., Wang, Z., Ba, S. & Hu, X. (2023) Preliminary exploration of the effectiveness of music listening and music recommender for studying in naturalistic settings. In Proceedings of the 23rd IEEE ICALT. (Best Student Paper Award)

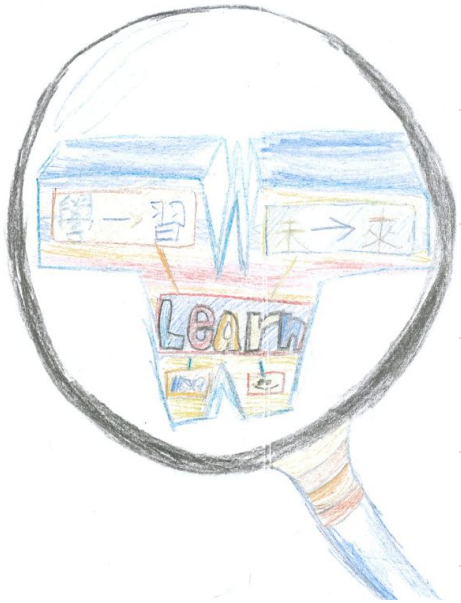


Take-away from Naturalistic Experiments

- Exploit multimodal data & their interactions
  - Physiological data w/ consumer devices
  - Behavioral logs
  - Explicit / implicit user feedback
- Can accommodate more scenarios of music use
  - Collaborative learning
  - Learning in virtual reality
  - Different learning tasks
- Fine tune recommendation models for various scenarios

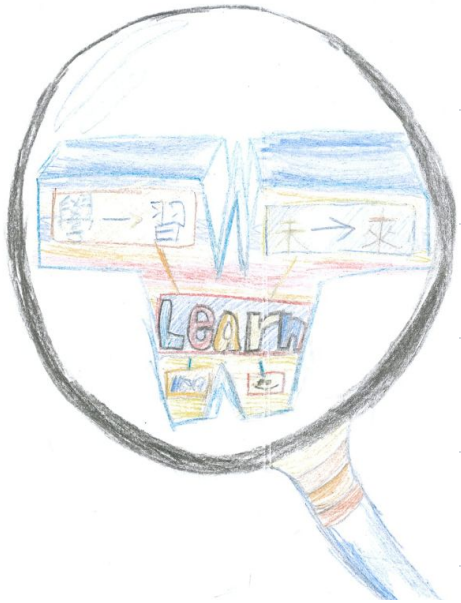


Agenda



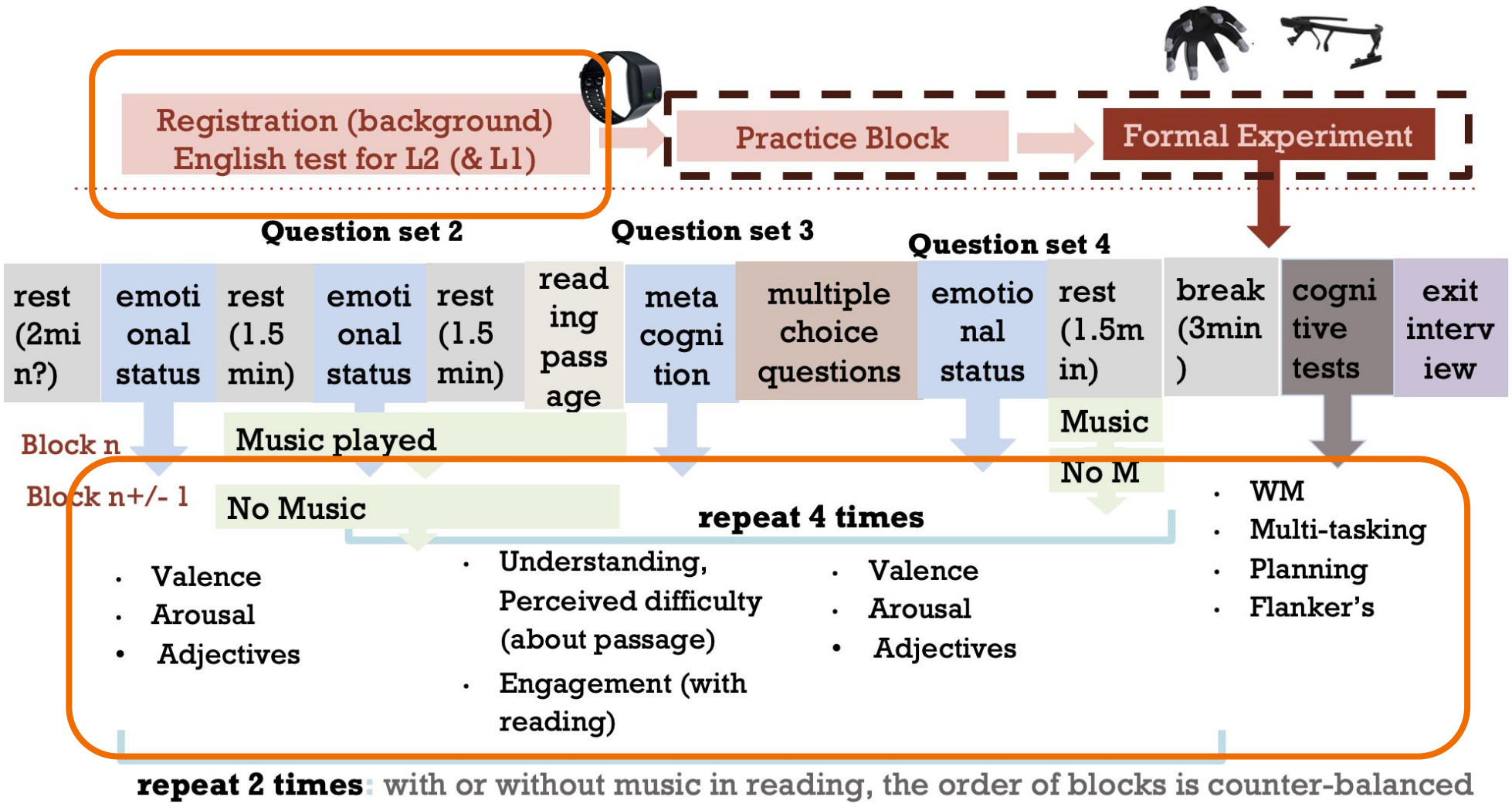
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14:00–14:10	1 Welcome & Self-introduction	Xiao Hu
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How to Measure the Users?



Data Collection Instruments

Pre-questionnaire

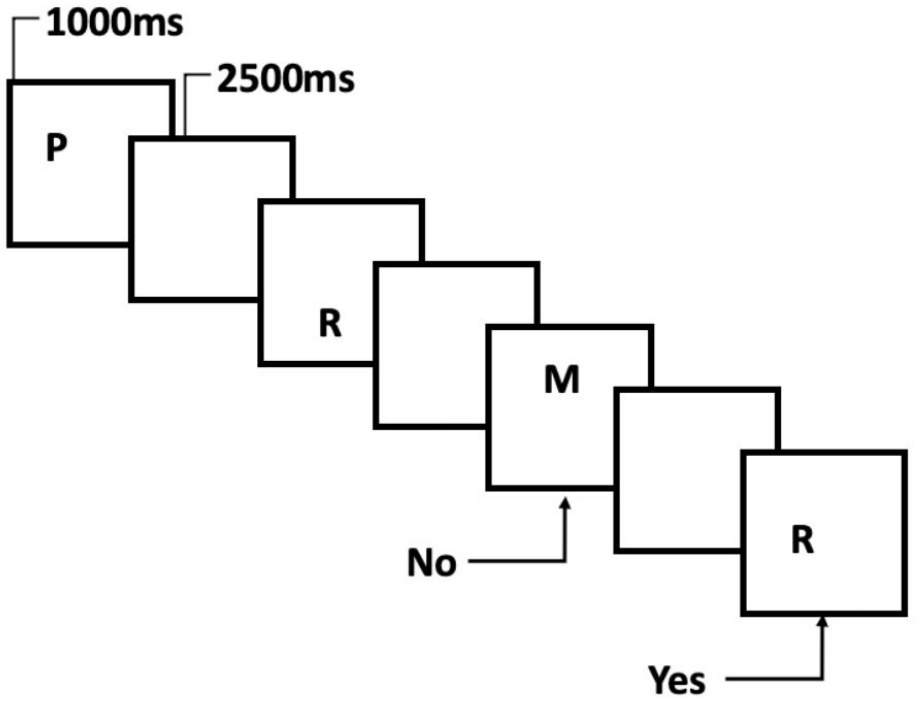
- Demographic
  - e.g., Gender, Age, Major
- Music Experience (Hu et al., 2019)
  - e.g., Music Training, Music Preference, BM Listening Frequency
- Personality Traits: Ten Item Personality Inventory (Romero et al., 2012)
  - e.g., degree of extraversion, agreeableness, conscientiousness, neuroticism, openness

Cognitive Ability Tests

- Working memory capacity by Two-back test (Lau et al., 2010)
- Multi-tasking ability by Dual-tasking Task (Stoet et al., 2013)
- Reading Task: English proficiency level measured by LexTable test (Lemhöfer & Broersma, 2012)

Working Memory Capacity

- The ability to maintain attention amid distractions
- Rule: Match the current trial stimulus from two trials earlier
- Type: Verbal = Match same letter; Spatial = Match same location
- Measure: Mean response time on correct responses
- Longer RT → lower WMC

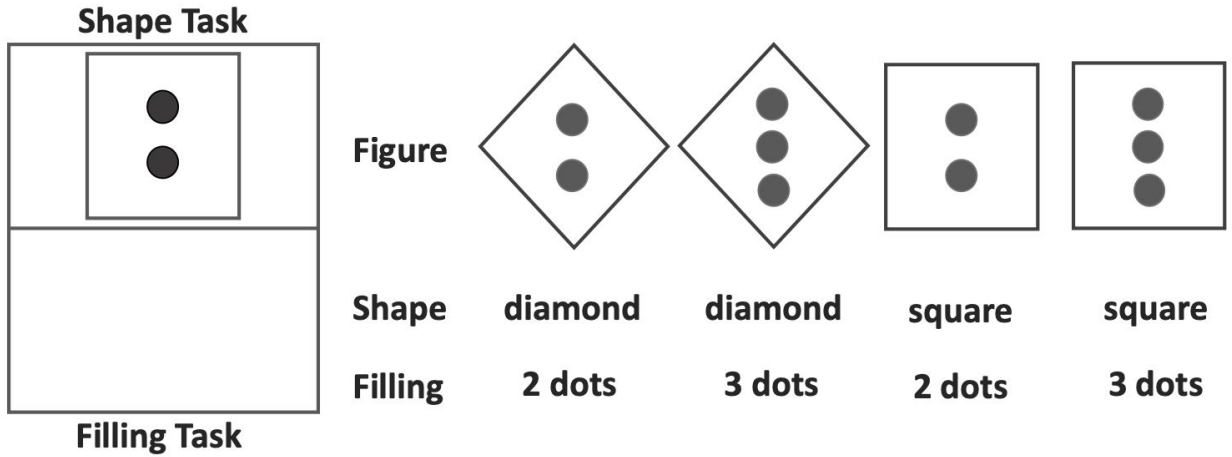


Two-back test (Lau et al., 2010)

Multitasking Ability

- Process multiple information streams while ignoring distractions
- Rule: Top position → Judge shape;

  Bottom position → Judge filling
- Single tasks = Only judge shape or filling; Dual task = Switch between two judgments based on position
- Measure: RT in dual task − RT across single tasks for correct-response
- Larger RT difference → lower multitasking



Dual-tasking Task (Stoet et al., 2013)

English Proficiency

- Rule: Decide whether each letter string is a valid English word
- Type: Word / Nonword judgment
- Measure: Percentage of correct responses across all trials
- Higher accuracy → higher English proficiency



LexTable test (Lemhöfer & Broersma, 2012)

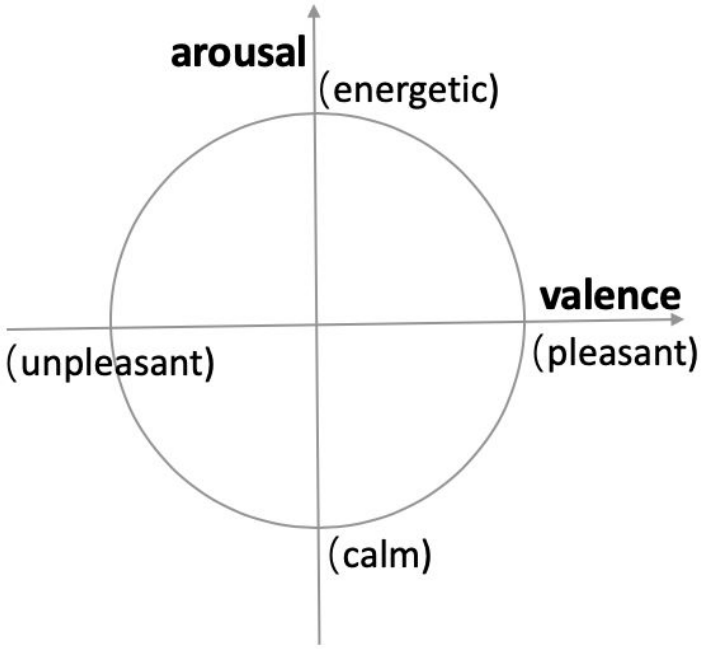
Interview Protocol

- Exit interviews performed to understand the reasons behind participants' opinions and thoughts on studying with background music.

- Example questions:
- (1) How do you like studying with music in the background? Why or Why not?
- (2) Which aspects of music have positive/negative influences on learning?
- (3) Music has positive/negative impacts on which aspects of your learning?
- (4) What do you think about this experiment? - Prompts: Procedure, Task, Question

Behavioural and Self-reported Measures

- Task Performance
  - Task Accuracy, Task Viewing Time, Question Answering Time
- Emotion
  - Change of Valence & Arousal (Russel, 1989)
  - Difference Before & After a task period or a resting period
  - Physiological signals
- Meta-cognition
  - How engaged are you during the task?
  - Do you think this task is difficult?
  - Do you think this task is interesting?
  - How satisfied do you perceive in the task?

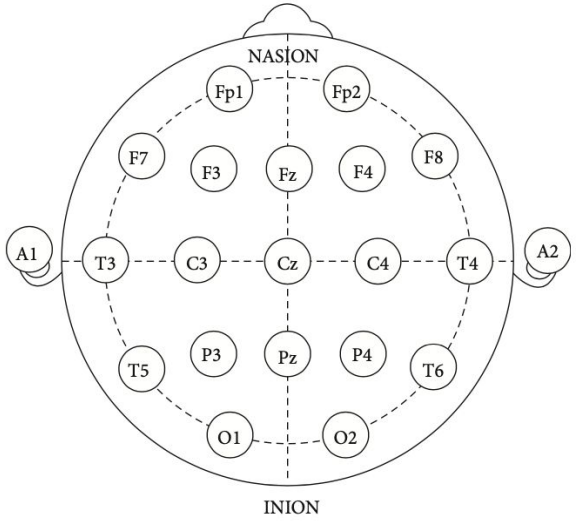


Multimodal Data Sources

- Emotion
  - Facial expression
  - EDA
  - Heart rate (variability)
  - Pupil dilation
  - Skin temperature
- Cognition
  - EEG
  - Eye movement
  - EDA
  - Heart rate (variability)

Measures: EEG Signals

- EEG features extracted from multiple frequency bands of our interests:
  - delta (1-4Hz), theta (4-8Hz), alpha (8-13Hz), lower alpha (8-10Hz), higher alpha (11-13Hz), beta (13-30Hz), beta 1 (12.5-16Hz), beta2 (16.5-20Hz), beta3 (20.5-28Hz), gamma (30-80Hz)
  - follow a method in (Groppe et al., 2013) with EEGLAB (Delorme et al., 2011).
  - compute mean log spectrum of action period from multiple channel locations.



EEG electrode positioning system

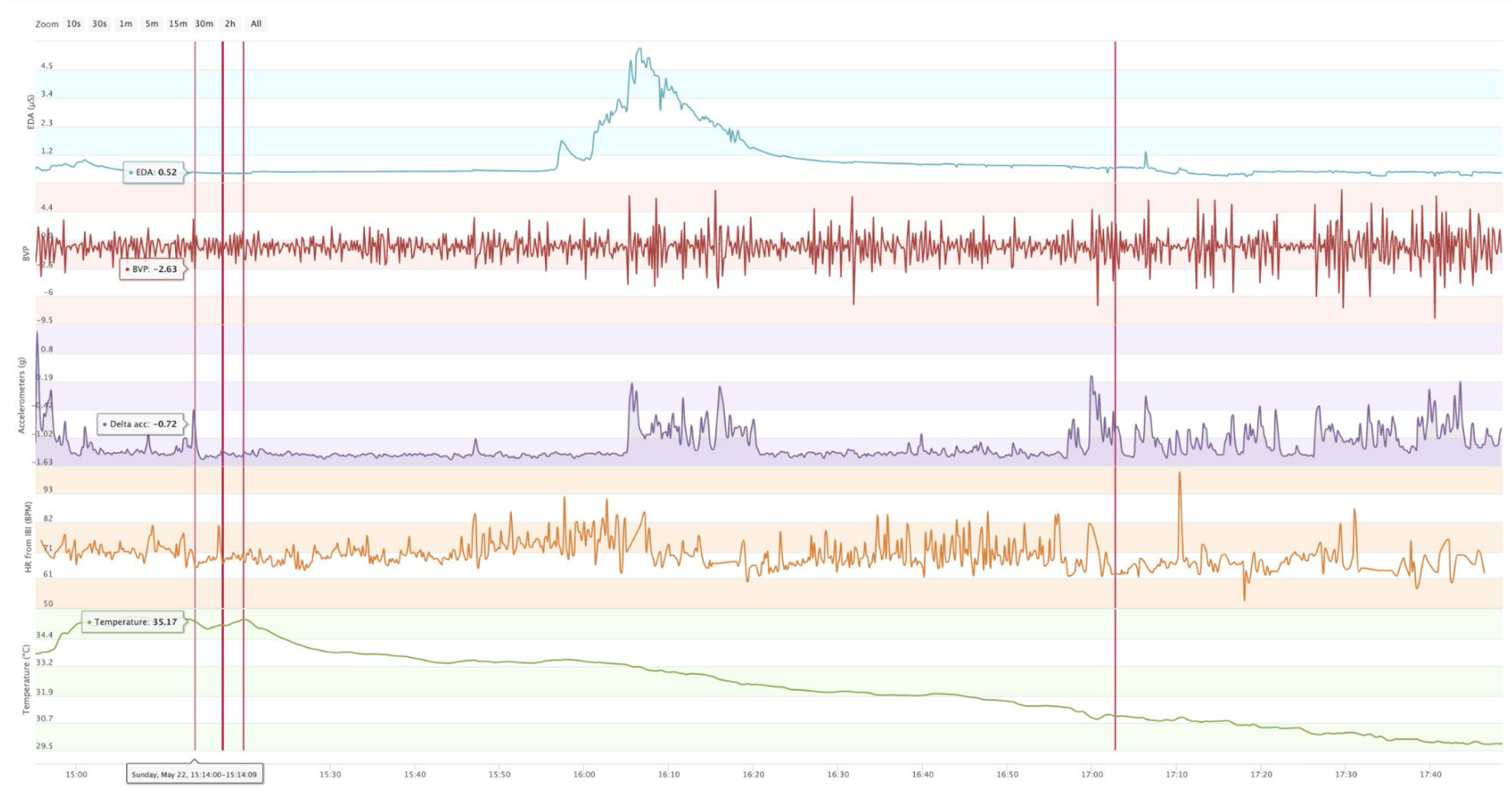
Measures: Eye Movements

- Classical eye-tracking measures (Rayner et al., 1998)
  - fixation duration/count
  - saccade duration/count/amplitude
  - regression count
  - skipping rate
  - pupil size change (No for VR experiment)

- HMM entropy (Chuk et al., 2014)
  - summarize eye movement pattern into HMM model (Hidden Markov Model)
  - contain personalized regions of interests (ROIs) & transition probabilities among ROIs
  - estimate entropy for each HMM model (Krejtz et al., 2014)
  - HMM entropy captures predictability and consistency of eye-movement patterns
  - Lower entropy can indicate a stable, well-practiced reading strategy, whereas higher entropy suggest dispersed attention

Physiological Data Collected

EDA BVP IBI HR TEMP



Measures: Peripheral Physiological Signals

Descriptive statistics

- Heart Rate, Blood Volume Pulse, Electrodermal Activity, Skin Temperature
- Mean, Median, Standard Deviation, Range (Liu, 2019)

Derived physiological measures from heart rate variability

- Time domain Features: SDSD, RMSSD, NN20, NN50, pNN20, pNN50, CVSD, CVNN (Hu et al., 2019)
- Frequency domain Features: Total power, VLF, LF, HF, LF/HF, LFNU, HFNU (Hu et al., 2019)

Derived physiological measures from electrodermal activity

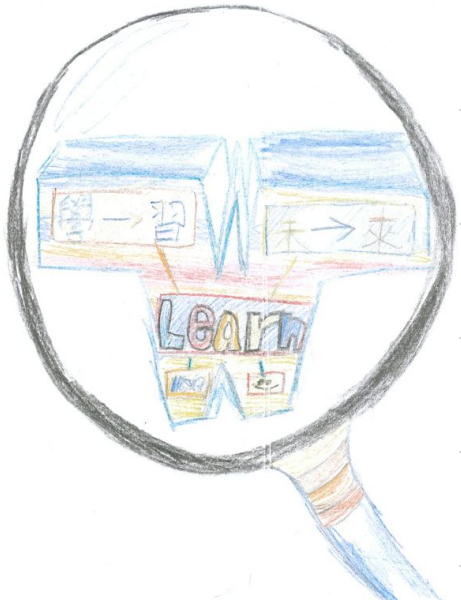
- SCR (e.g., frequency, amplitude) (Hu et al., 2022; Que et al., 2024)
- SCL (Hu et al., 2022)

Measures: Music Characteristics

Extract low-level (acoustic features) & high-level (genre, lyric, emotion) features for each piece.

- Acoustic Features (i.e., statistics elicited from audio signals of music) extracted by Librosa and other tools in MIR
  - Rhythm (tempo, rhythm strength)
  - Loudness (RMSE)
  - Timbre (centroid, flatness, roll-off)
  - Mode (major vs minor)
- Emotion
  - contain two dimensions of happiness and energy (Russel, 1989)
  - predicted based on acoustic features with classification algorithm, e.g., SVM (Hu & Yang, 2017)
- Genre (e.g., classical, Jazz) were classified using machine learning models
- Genre and the presence of lyrics can be manually coded for small datasets
  - Interrater reliability will be computed with Kappa statistic

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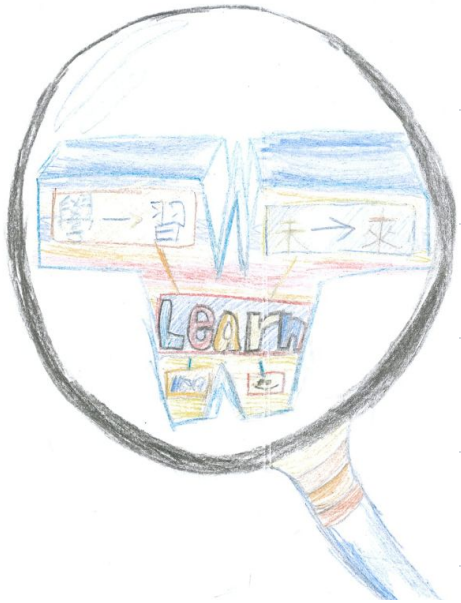
Hands-on demo



- Google Colab Notebook with sample data
  - https://bit.ly/m4ldemo

- Content
  - Physiological signal processing – 15 mins
  - Naturalistic experiment data analysis – 15 mins

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(Some) Open challenges for MIR and learning

- **Evaluation:** What counts as helpful music?
- **Ecology:** How can we balance learner choice and experimental control?
- **Generalization:** Which findings transfer across other learning situations?
- **Ethics:** When is it worth adding eye tracking, EDA, or EEG, given the extra effort and privacy concerns?

Recap of Today's Objectives

- A methodological framework for human-centered research on music for learning
- Integration of music features with behavioral, subjective, and physiological measures
- How to measure listeners' cognitive processes and affective responses to music
- Practical considerations for experimental design, multimodal data collection, measurement, and modelling

Quick Reflections

Please fill in: https://bit.ly/M4L_Padlet

- Which learning objectives have you obtained?
  - Methodological framework
  - Integration of measures
  - How to measure
  - Practical considerations



- What is the one thing you'll try in your research?

- What topic(s) would you like to know more?

Thank You!

Feel free to contact us at

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laualan@connect.hku.hk

for any inquiries or collaboration.

https://ccmirlab.com/

https://www.researchgate.net/lab/CCMIR-Xiao-Hu